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Refrigerant vessel and refrigeration cycle apparatus

Abstract

Provided are a refrigerant vessel and a refrigeration cycle apparatus capable of suppressing accumulation of a refrigerating machine oil inside. A secondary-side receiver reserves a refrigerant in a secondary-side refrigerant circuit in which a refrigerant and a refrigerating machine oil circulate, the secondary-side receiver including a vessel body, a first refrigerant pipe connected to the vessel body, and a second refrigerant pipe connected to the vessel body, in which the refrigerating machine oil is incompatible with the refrigerant and has a higher density than the refrigerant, and the second refrigerant pipe extends downward from a bottom part of the vessel body.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of PCT International Application No. PCT/JP2022/035473, filed on Sep. 22, 2022, which claims priority under 35 U.S.C. § 119(a) to Patent Application No. JP 2021-161999, filed in Japan on Sep. 30, 2021, all of which are hereby expressly incorporated by reference into the present application.

(1) The present disclosure relates to a refrigerant vessel and a refrigeration cycle apparatus.

TECHNICAL FIELD

Background Art

(2) Conventionally, a receiver for reserving a refrigerant has been used in a refrigerant circuit included in a refrigeration cycle apparatus.

(3) As such a refrigeration cycle apparatus, for example, a refrigeration cycle apparatus described in Patent Literature 1 (JP 2008-164225 A) uses a carbon dioxide refrigerant as a refrigerant and uses incompatible PAG oil as a refrigerating machine oil. With this refrigeration cycle apparatus, in order to avoid accumulation of refrigerating machine oil at a bottom of a refrigerant vessel such as a receiver, it is proposed that an outlet pipe connected to a side peripheral surface of the refrigerant vessel is disposed so that a vicinity of a distal end located inside the refrigerant vessel is obliquely cut and the distal end is in contact with a bottom part of the refrigerant vessel.

SUMMARY

(4) A refrigerant vessel according to a first aspect is a refrigerant vessel that reserves a refrigerant in a refrigerant circuit. In the refrigerant circuit, the refrigerant and the refrigerating machine oil circulate. The refrigerant vessel includes a vessel body, a first refrigerant pipe, and a second refrigerant pipe. The first refrigerant pipe is connected to the vessel body. The second refrigerant

pipe is connected to the vessel body. The refrigerating machine oil is incompatible with the refrigerant. The refrigerating machine oil has a higher density than the refrigerant. The second refrigerant pipe extends downward from a bottom part of the vessel body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic configuration diagram of a refrigeration cycle apparatus.
- (2) FIG. 2 is a schematic functional block configuration diagram of the refrigeration cycle apparatus.
- (3) FIG. 3 is a diagram illustrating motion (a flow of a refrigerant) in cooling operation of the refrigeration cycle apparatus.
- (4) FIG. 4 is a diagram illustrating motion (a flow of a refrigerant) in heating operation of the refrigeration cycle apparatus.
- (5) FIG. 5 is a diagram illustrating motion (a flow of a refrigerant) in simultaneous cooling and heating operation (cooling main operation) of the refrigeration cycle apparatus.
- (6) FIG. 6 is a diagram illustrating motion (a flow of a refrigerant) in simultaneous cooling and heating operation (heating main operation) of the refrigeration cycle apparatus.
- (7) FIG. 7 is a schematic configuration diagram of a secondary-side receiver.
- (8) FIG. 8 is a schematic configuration diagram of a secondary-side receiver according to another embodiment A.
- (9) FIG. 9 is a schematic configuration diagram of a secondary-side receiver according to another embodiment B.
- (10) FIG. 10 is a schematic configuration diagram of a secondary-side receiver according to another embodiment C.
- (11) FIG. 11 is a schematic configuration diagram of a secondary-side receiver according to another embodiment D.
- (12) FIG. 12 is a schematic configuration diagram of a refrigerant circuit including an intermediate-pressure receiver according to another embodiment E.
- (13) FIG. 13 is a schematic configuration diagram of a refrigeration cycle apparatus according to another embodiment F.

DESCRIPTION OF EMBODIMENTS

(1) Configuration of Refrigeration Cycle Apparatus

- (14) FIG. 1 is a schematic configuration diagram of a refrigeration cycle apparatus 1. FIG. 2 is a schematic functional block configuration diagram of the refrigeration cycle apparatus 1.
- (15) The refrigeration cycle apparatus 1 is an apparatus used for cooling and heating of an indoor space in a building or the like by performing vapor compression refrigeration cycle operation.
- (16) The refrigeration cycle apparatus 1 includes a binary refrigerant circuit including a vapor compression primary-side refrigerant circuit 5a (corresponding to a first refrigerant circuit) and a vapor compression secondary-side refrigerant circuit 10 (corresponding to a refrigerant circuit, a second refrigerant circuit), and performs a binary refrigeration cycle. In the present embodiment, for example, R32 or R410A is sealed as a refrigerant in the primary-side refrigerant circuit 5a. In the secondary-side refrigerant circuit 10, for example, carbon dioxide is sealed as a refrigerant. In the secondary-side refrigerant circuit 10, a refrigerating machine oil incompatible with the refrigerant is used as a refrigerating machine oil that circulates in the circuit together with the refrigerant to improve lubricity of a sliding portion and the like. The refrigerating machine oil has a higher density and a higher specific gravity than the refrigerant in a use state, and thus tends to be located below the refrigerant. In the present embodiment, a polyalkylene glycol oil (PAG oil) that is incompatible with carbon dioxide as a refrigerant and has a higher density than the carbon dioxide

refrigerant is used.

(17) The primary-side refrigerant circuit **5a** and the secondary-side refrigerant circuit **10** are thermally connected via a cascade heat exchanger **35** described later.

(18) The refrigeration cycle apparatus **1** is configured by connecting a primary-side unit **5**, a cascade unit **2**, a plurality of branch units **6a**, **6b**, and **6c**, and a plurality of utilization units **3a**, **3b**, and **3c** to each other via pipes. The primary-side unit **5** and the cascade unit **2** are connected via a primary-side first connection pipe **111** and a primary-side second connection pipe **112**. The cascade unit **2** and the plurality of branch units **6a**, **6b**, and **6c** are connected via three connection pipes, namely, a secondary-side second connection pipe **9**, a secondary-side first connection pipe **8**, and a secondary-side third connection pipe **7**. The plurality of branch units **6a**, **6b**, and **6c** and the plurality of utilization units **3a**, **3b**, and **3c** are connected via first connecting tubes **15a**, **15b**, and **15c** and second connecting tubes **16a**, **16b**, and **16c**. The present embodiment provides the single primary-side unit **5**. The present embodiment provides the single cascade unit **2**. The plurality of utilization units **3a**, **3b**, and **3c** according to the present embodiment includes three utilization units, namely, the first utilization unit **3a**, the second utilization unit **3b**, and the third utilization unit **3c**. In the present embodiment, the plurality of branch units **6a**, **6b**, and **6c** is three branch units of the first branch unit **6a**, the second branch unit **6b**, and the third branch unit **6c**.

(19) In the refrigeration cycle apparatus **1**, the utilization units **3a**, **3b**, and **3c** can individually perform cooling operation or heating operation, and heat can be recovered between the utilization units by sending a refrigerant from the utilization unit performing the heating operation to the utilization unit performing the cooling operation. Specifically, heat is recovered in the present embodiment by performing cooling main operation or heating main operation of simultaneously performing cooling operation and heating operation. In addition, the refrigeration cycle apparatus **1** is configured to balance heat loads of the cascade unit **2** in accordance with entire heat loads of the plurality of utilization units **3a**, **3b**, and **3c** also in consideration of the heat recovery (the cooling main operation or the heating main operation).

(2) Primary-Side Refrigerant Circuit

(20) The primary-side refrigerant circuit **5a** includes a primary-side compressor **71**, a primary-side switching mechanism **72**, a primary-side heat exchanger **74**, a primary-side first expansion valve **76**, a primary-side subcooling heat exchanger **103**, a primary-side subcooling circuit **104**, a primary-side subcooling expansion valve **104a**, a first liquid shutoff valve **108**, the primary-side first connection pipe **111**, a second liquid shutoff valve **106**, the second refrigerant pipe **114**, a primary-side second expansion valve **102**, the cascade heat exchanger **35** shared with the secondary-side refrigerant circuit **10**, a first refrigerant pipe **113**, a second gas shutoff valve **107**, the primary-side second connection pipe **112**, a first gas shutoff valve **109**, and a primary-side accumulator **105**. This primary-side refrigerant circuit **5a** specifically includes a primary-side flow path **35b** of the cascade heat exchanger **35**.

(21) The primary-side compressor **71** is a device for compressing a primary-side refrigerant, and includes, for example, a scroll type or other positive-displacement compressor whose operating capacity can be varied by controlling an inverter for a compressor motor **71a**.

(22) The primary-side accumulator **105** is provided at a halfway portion of the suction flow path connecting the primary-side switching mechanism **72** and a suction side of the primary-side compressor **71**.

(23) When the cascade heat exchanger **35** functions as an evaporator for the primary-side refrigerant, the primary-side switching mechanism **72** enters a fifth connection state of connecting the suction side of the primary-side compressor **71** and a gas side of a primary-side flow path **35b** of the cascade heat exchanger **35** (see solid lines in the primary-side switching mechanism **72** in FIG. 1). In addition, when the cascade heat exchanger **35** functions as a radiator for the primary-side refrigerant, the primary-side switching mechanism **72** enters a sixth connection state of connecting a discharge side of the primary-side compressor **71** and the gas side of the primary-side

flow path **35b** of the cascade heat exchanger **35** (see broken lines in the primary-side switching mechanism **72** in FIG. **1**). In such a manner, the primary-side switching mechanism **72** is a device that can switch the flow path of the refrigerant in the primary-side refrigerant circuit **5a**, and includes, for example, a four-way switching valve. Then, by changing a switching state of the primary-side switching mechanism **72**, the cascade heat exchanger **35** can function as the evaporator or the radiator for the primary-side refrigerant.

(24) The cascade heat exchanger **35** is a device for causing heat exchange between a refrigerant such as R32 which is a primary-side refrigerant and a refrigerant such as carbon dioxide which is a secondary-side refrigerant without mixing the refrigerants with each other. The cascade heat exchanger **35** is, for example, a plate-type heat exchanger. The cascade heat exchanger **35** includes a secondary-side flow path **35a** belonging to the secondary-side refrigerant circuit **10** and the primary-side flow path **35b** belonging to the primary-side refrigerant circuit **5a**. The secondary-side flow path **35a** has a gas side connected to a secondary-side switching mechanism **22** via a third pipe **25**, and a liquid side connected to a cascade expansion valve **36** via a fourth pipe **26**. The primary-side flow path **35b** has a gas side connected to the primary-side compressor **71** via the first refrigerant pipe **113**, the second gas shutoff valve **107**, the primary-side second connection pipe **112**, the first gas shutoff valve **109**, and the primary-side switching mechanism **72**, and has a liquid side connected to the second refrigerant pipe **114** provided with the primary-side second expansion valve **102**.

(25) The primary-side heat exchanger **74** is a device for exchanging heat between the primary-side refrigerant and outdoor air. The primary-side heat exchanger **74** has a gas side connected to a pipe extending from the primary-side switching mechanism **72**. The primary-side heat exchanger **74** includes, for example, a fin-and-tube heat exchanger including a large number of heat transfer tubes and fins.

(26) The primary-side first expansion valve **76** is provided on a liquid pipe extending from a liquid side of the primary-side heat exchanger **74** to the primary-side subcooling heat exchanger **103**. The primary-side first expansion valve **76** is an electrically powered expansion valve that has an adjustable opening degree for adjusting a flow rate of the primary-side refrigerant flowing in a portion on a liquid side of the primary-side refrigerant circuit **5a**.

(27) The primary-side subcooling circuit **104** branches from a portion between the primary-side first expansion valve **76** and the primary-side subcooling heat exchanger **103**, and is connected to a portion between the primary-side switching mechanism **72** and the primary-side accumulator **105** on the suction flow path. The primary-side subcooling expansion valve **104a** is an electrically powered expansion valve that is provided upstream of the primary-side subcooling heat exchanger **103** in the primary-side subcooling circuit **104** and has an adjustable opening degree for adjusting the flow rate of the primary-side refrigerant.

(28) The primary-side subcooling heat exchanger **103** causes heat exchange between a refrigerant flowing from the primary-side first expansion valve **76** toward the first liquid shutoff valve **108** and a refrigerant decompressed at the primary-side subcooling expansion valve **104a** in the primary-side subcooling circuit **104**.

(29) The primary-side first connection pipe **111** is a pipe connecting the first liquid shutoff valve **108** and the second liquid shutoff valve **106**, and connects the primary-side unit **5** and the cascade unit **2**.

(30) The primary-side second connection pipe **112** is a pipe connecting the first gas shutoff valve **109** and the second gas shutoff valve **107**, and connects the primary-side unit **5** and the cascade unit **2**.

(31) The second refrigerant pipe **114** is a pipe extending from a liquid side of the primary-side flow path **35b** of the cascade heat exchanger **35** to the second liquid shutoff valve **106**.

(32) The primary-side second expansion valve **102** is provided on the second refrigerant pipe **114**. The primary-side second expansion valve **102** is an electric expansion valve that has an adjustable

opening degree for adjusting a flow rate of the primary-side refrigerant flowing in the primary-side flow path **35b** of the cascade heat exchanger **35**.

(33) The first refrigerant pipe **113** is a pipe extending from the gas side of the primary-side flow path **35b** of the cascade heat exchanger **35** to the second gas shutoff valve **107**.

(34) The first gas shutoff valve **109** is provided between the primary-side second connection pipe **112** and the primary-side switching mechanism **72**.

(3) Secondary-Side Refrigerant Circuit

(35) The secondary-side refrigerant circuit **10** includes the plurality of utilization units **3a**, **3b**, and **3c**, the plurality of branch units **6a**, **6b**, and **6c**, and the cascade unit **2**, which are connected to each other. Each of the utilization units **3a**, **3b**, and **3c** is connected to a corresponding one of the branch units **6a**, **6b**, and **6c** one on one. Specifically, the utilization unit **3a** and the branch unit **6a** are connected via the first connecting tube **15a** and the second connecting tube **16a**, the utilization unit **3b** and the branch unit **6b** are connected via the first connecting tube **15b** and the second connecting tube **16b**, and the utilization unit **3c** and the branch unit **6c** are connected via the first connecting tube **15c** and the second connecting tube **16c**. Each of the branch units **6a**, **6b**, and **6c** are connected to the cascade unit **2** via three connection pipes, namely, the secondary-side third connection pipe **7**, the secondary-side first connection pipe **8**, and the secondary-side second connection pipe **9**. Specifically, the secondary-side third connection pipe **7**, the secondary-side first connection pipe **8**, and the secondary-side second connection pipe **9** extending from the cascade unit **2** are each branched into a plurality of pipes connected to the branch units **6a**, **6b**, and **6c**.

(36) In accordance with an operating state, either the refrigerant in a gas-liquid two-phase state or the refrigerant in a gas state flows in the secondary-side first connection pipe **8**. Note that, in accordance with the operating state, the refrigerant in a supercritical state flows in the secondary-side first connection pipe **8**. In accordance with the operating state, either the refrigerant in the gas-liquid two-phase state or the refrigerant in the gas state flows in the secondary-side second connection pipe **9**. In accordance with the operating state, either the refrigerant in the gas-liquid two-phase state or the refrigerant in a liquid state flows in the secondary-side third connection pipe **7**. Note that, in accordance with the operating state, the refrigerant in a supercritical state flows in the secondary-side third connection pipe **7**.

(37) The secondary-side refrigerant circuit **10** includes a cascade circuit **12**, branch circuits **14a**, **14b**, and **14c**, and utilization circuits **13a**, **13b**, and **13c**, which are connected to each other.

(38) The cascade circuit **12** principally includes a secondary-side compressor **21**, the secondary-side switching mechanism **22**, a first pipe **28**, a second pipe **29**, a suction flow path **23**, a discharge flow path **24**, the third pipe **25**, the fourth pipe **26**, a fifth pipe **27**, the cascade heat exchanger **35**, the cascade expansion valve **36**, a third shutoff valve **31**, a first shutoff valve **32**, a second shutoff valve **33**, a secondary-side accumulator **30**, an oil separator **34**, an oil return circuit **40**, a secondary-side receiver **45** (corresponding to a refrigerant vessel), a bypass circuit **46**, a bypass expansion valve **46a**, a secondary-side subcooling heat exchanger **47**, a secondary-side subcooling circuit **48**, and a secondary-side subcooling expansion valve **48a**. The cascade circuit **12** of the secondary-side refrigerant circuit **10** specifically includes the secondary-side flow path **35a** of the cascade heat exchanger **35**.

(39) The secondary-side compressor **21** is a device for compressing the secondary-side refrigerant, and is exemplarily constituted by a scroll type or other positive-displacement compressor whose operating capacity can be varied by controlling an inverter for a compressor motor **21a**. The secondary-side compressor **21** is controlled in accordance with an operating load so as to have larger operating capacity as the load increases.

(40) The secondary-side switching mechanism **22** is a mechanism that can switch a connection state of the secondary-side refrigerant circuit **10**, specifically, the flow path of the refrigerant in the cascade circuit **12**. In the present embodiment, the secondary-side switching mechanism **22** includes a discharge-side connection portion **22x**, a suction-side connection portion **22y**, a first

switching valve **22a**, and a second switching valve **22b**. An end of the discharge flow path **24** on a side opposite to the secondary-side compressor **21** is connected to the discharge-side connection portion **22x**. An end of the suction flow path **23** on a side opposite to the secondary-side compressor **21** is connected to the suction-side connection portion **22y**. The first switching valve **22a** and the second switching valve **22b** are provided in parallel to each other between the discharge flow path **24** and the suction flow path **23** of the secondary-side compressor **21**. The first switching valve **22a** is connected to one end of the discharge-side connection portion **22x** and one end of the suction-side connection portion **22y**. The second switching valve **22b** is connected to the other end of the discharge-side connection portion **22x** and the other end of the suction-side connection portion **22y**. In the present embodiment, each of the first switching valve **22a** and the second switching valve **22b** includes a four-way switching valve. Each of the first switching valve **22a** and the second switching valve **22b** has four connection ports, namely, a first connection port, a second connection port, a third connection port, and a fourth connection port. In the first switching valve **22a** and the second switching valve **22b** according to the present embodiment, each of the fourth ports is a closed connection port not connected to the flow path of the secondary-side refrigerant circuit **10**. In the first switching valve **22a**, the first connection port is connected to the one end of the discharge-side connection portion **22x**, the second connection port is connected to the third pipe **25** extending from the secondary-side flow path **35a** of the cascade heat exchanger **35**, and the third connection port is connected to the one end of the suction-side connection portion **22y**. The first switching valve **22a** switches between a switching state in which the first connection port and the second connection port are connected and the third connection port and the fourth connection port are connected and a switching state in which the third connection port and the second connection port are connected and the first connection port and the fourth connection port are connected. The second switching valve **22b** has the first connection port connected to the other end of the discharge-side connection portion **22x**, the second connection port connected to the first pipe **28**, and the third connection port connected to the other end of the suction-side connection portion **22y**. The second switching valve **22b** switches between a switching state in which the first connection port and the second connection port are connected and the third connection port and the fourth connection port are connected and a switching state in which the third connection port and the second connection port are connected and the first connection port and the fourth connection port are connected.

(41) When the secondary-side refrigerant discharged from the secondary-side compressor **21** is prevented from being sent to the secondary-side first connection pipe **8** while the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant, the secondary-side switching mechanism **22** is switched to a first connection state in which the discharge flow path **24** and the third pipe **25** are connected by the first switching valve **22a** and the first pipe **28** and the suction flow path **23** are connected by the second switching valve **22b**. The first connection state of the secondary-side switching mechanism **22** is a connection state adopted during the cooling operation described later. When the cascade heat exchanger **35** functions as an evaporator for the secondary-side refrigerant, the secondary-side switching mechanism **22** is switched to a second connection state in which the discharge flow path **24** and the first pipe **28** are connected by the second switching valve **22b** and the third pipe **25** and the suction flow path **23** are connected by the first switching valve **22a**. The second connection state of the secondary-side switching mechanism **22** is a connection state adopted during the heating operation and during the heating main operation described later. When the secondary-side refrigerant discharged from the secondary-side compressor **21** is sent to the secondary-side first connection pipe **8** while the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant, the secondary-side switching mechanism **22** is switched to a third connection state in which the discharge flow path **24** and the third pipe **25** are connected by the first switching valve **22a** and the discharge flow path **24** and the first pipe **28** are connected by the second switching valve **22b**. The third connection state of

the secondary-side switching mechanism **22** is a connection state adopted during the cooling main operation described later.

(42) As described above, the cascade heat exchanger **35** is a device for causing heat exchange between the refrigerant such as R32 which is the primary-side refrigerant and the refrigerant such as carbon dioxide which is the secondary-side refrigerant without mixing the refrigerants with each other. The cascade heat exchanger **35** includes the secondary-side flow path **35a** in which the secondary-side refrigerant in the secondary-side refrigerant circuit **10** flows and the primary-side flow path **35b** in which the primary-side refrigerant in the primary-side refrigerant circuit **5a** flows, so as to be shared between the primary-side unit **5** and the cascade unit **2**. Note that, in the present embodiment, the cascade heat exchanger **35** is disposed inside a cascade casing (not illustrated) of the cascade unit **2**. The gas side of the primary-side flow path **35b** of the cascade heat exchanger **35** extends to the primary-side second connection pipe **112** outside the cascade casing via the first refrigerant pipe **113** and the second gas shutoff valve **107**. The liquid side of the primary-side flow path **35b** of the cascade heat exchanger **35** extends to the primary-side first connection pipe **111** outside the cascade casing via the second refrigerant pipe **114** provided with the primary-side second expansion valve **102** and the second liquid shutoff valve **106**.

(43) The cascade expansion valve **36** is an expansion valve for adjusting a flow rate of the secondary-side refrigerant flowing in the cascade heat exchanger **35**. The cascade expansion valve **36** is an electric expansion valve that is connected to a liquid side of the cascade heat exchanger **35** and has an adjustable opening degree. The cascade expansion valve **36** is provided on the fourth pipe **26**.

(44) Each of the third shutoff valve **31**, the first shutoff valve **32**, and the second shutoff valve **33** is provided at a connecting port with an external device or pipe (specifically, the connection pipe **7**, **8**, or **9**). Specifically, the third shutoff valve **31** is connected to the secondary-side third connection pipe **7** led out of the cascade unit **2**. The first shutoff valve **32** is connected to the secondary-side first connection pipe **8** led out of the cascade unit **2**. The second shutoff valve **33** is connected to the secondary-side second connection pipe **9** led out of the cascade unit **2**.

(45) The first pipe **28** is a refrigerant pipe connecting the first shutoff valve **32** and the secondary-side switching mechanism **22**. Specifically, the first pipe **28** connects the first shutoff valve **32** and the second connection port of the second switching valve **22b** of the secondary-side switching mechanism **22**.

(46) The suction flow path **23** connects the secondary-side switching mechanism **22** and a suction side of the secondary-side compressor **21**. Specifically, the suction flow path **23** connects the suction-side connection portion **22y** of the secondary-side switching mechanism **22** and the suction side of the secondary-side compressor **21**. The secondary-side accumulator **30** is provided at a halfway portion of the suction flow path **23**.

(47) The second pipe **29** is a refrigerant pipe that connects the second shutoff valve **33** to a halfway portion of the suction flow path **23**. In the present embodiment, the second pipe **29** is connected to the suction flow path **23** at a connection point of the suction flow path **23** between the suction-side connection portion **22y** of the secondary-side switching mechanism **22** and the secondary-side accumulator **30**.

(48) The discharge flow path **24** is a refrigerant pipe connecting a discharge side of the secondary-side compressor **21** and the secondary-side switching mechanism **22**. Specifically, the discharge flow path **24** connects the discharge side of the secondary-side compressor **21** and the discharge-side connection portion **22x** of the secondary-side switching mechanism **22**.

(49) The third pipe **25** is a refrigerant pipe connecting the secondary-side switching mechanism **22** and a gas side of the cascade heat exchanger **35**. Specifically, the third pipe **25** connects the second connection port of the first switching valve **22a** of the secondary-side switching mechanism **22** and a gas-side end of the secondary-side flow path **35a** in the cascade heat exchanger **35**.

(50) The fourth pipe **26** is a refrigerant pipe connecting the liquid side (opposite to the gas side, and

opposite to the side provided with the secondary-side switching mechanism 22) of the cascade heat exchanger 35 and the secondary-side receiver 45. Specifically, the fourth pipe 26 connects a liquid side end (opposite to the gas side) of the secondary-side flow path 35a in the cascade heat exchanger 35 and the secondary-side receiver 45.

(51) The secondary-side receiver 45 is a refrigerant vessel that reserves a residue refrigerant in the secondary-side refrigerant circuit 10. Extended from the secondary-side receiver 45 are the fourth pipe 26, the fifth pipe 27, and the bypass circuit 46.

(52) The bypass circuit 46 is a refrigerant pipe connecting a gas phase region which is an upper region in the secondary-side receiver 45 and the suction flow path 23. Specifically, the bypass circuit 46 is connected between the secondary-side switching mechanism 22 and the secondary-side accumulator 30 on the suction flow path 23. The bypass circuit 46 is provided with the bypass expansion valve 46a. The bypass expansion valve 46a is an electrically powered expansion valve that can adjust a quantity of the refrigerant guided from inside the secondary-side receiver 45 to the suction side of the secondary-side compressor 21 by adjusting an opening degree.

(53) The fifth pipe 27 is a refrigerant pipe connecting the secondary-side receiver 45 and the third shutoff valve 31.

(54) The secondary-side subcooling circuit 48 is a refrigerant pipe connecting a part of the fifth pipe 27 and the suction flow path 23. Specifically, the secondary-side subcooling circuit 48 is connected between the secondary-side switching mechanism 22 and the secondary-side accumulator 30 on the suction flow path 23. In the present embodiment, the secondary-side subcooling circuit 48 extends to branch from a portion between the secondary-side receiver 45 and the secondary-side subcooling heat exchanger 47.

(55) The secondary-side subcooling heat exchanger 47 is a heat exchanger that causes heat exchange between the refrigerant flowing in a flow path belonging to the fifth pipe 27 and the refrigerant flowing in a flow path belonging to the secondary-side subcooling circuit 48. In the present embodiment, the secondary-side subcooling heat exchanger 47 is provided between a portion from where the secondary-side subcooling circuit 48 branches and the third shutoff valve 31 on the fifth pipe 27. The secondary-side subcooling expansion valve 48a is provided between a portion branching from the fifth pipe 27 and the secondary-side subcooling heat exchanger 47 on the secondary-side subcooling circuit 48. The secondary-side subcooling expansion valve 48a is an electrically powered expansion valve that has an adjustable opening degree and supplies the secondary-side subcooling heat exchanger 47 with a decompressed refrigerant.

(56) The secondary-side accumulator 30 is a vessel that can reserve the secondary-side refrigerant, and is provided on the suction side of the secondary-side compressor 21.

(57) The oil separator 34 is provided at a halfway portion of the discharge flow path 24. The oil separator 34 is a device for separating refrigerating machine oil discharged from the secondary-side compressor 21 along with the secondary-side refrigerant from the secondary-side refrigerant and return the refrigerating machine oil to the secondary-side compressor 21.

(58) The oil return circuit 40 is provided to connect the oil separator 34 and the suction flow path 23. The oil return circuit 40 includes an oil return flow path 41 which is a flow path extending from the oil separator 34 and extending to join a portion between the secondary-side accumulator 30 and the suction side of the secondary-side compressor 21 on the suction flow path 23. An oil return capillary tube 42 and an oil return on-off valve 44 are provided at a halfway portion of the oil return flow path 41. When the oil return on-off valve 44 is controlled into an opened state, the refrigerating machine oil separated in the oil separator 34 passes through the oil return capillary tube 42 on the oil return flow path 41 and is returned to the suction side of the secondary-side compressor 21. In the present embodiment, when the secondary-side compressor 21 is in an operating state on the secondary-side refrigerant circuit 10, the oil return on-off valve 44 is kept in the opened state for predetermined time and is kept in a closed state for predetermined time repetitively, to control a returned quantity of the refrigerating machine oil through the oil return

circuit **40**. In the present embodiment, the oil return on-off valve **44** is an electromagnetic valve controlled to be opened and closed. Alternatively, the oil return on-off valve **44** may be an electrically powered expansion valve having an adjustable opening degree and not provided with the oil return capillary tube **42**.

(59) Hereinafter, the utilization circuits **13a**, **13b**, and **13c** will be described. Since the utilization circuits **13b** and **13c** are configured similarly to the utilization circuit **13a**, elements of the utilization circuits **13b** and **13c** will not be described repeatedly, assuming that a subscript “b” or “c” will replace a subscript “a” in reference signs denoting elements of the utilization circuit **13a**.

(60) The utilization circuit **13a** principally includes a utilization-side heat exchanger **52a**, a first utilization pipe **57a**, a second utilization pipe **56a**, and a utilization-side expansion valve **51a**.

(61) The utilization-side heat exchanger **52a** is a device for causing heat exchange between a refrigerant and indoor air, and includes, for example, a fin-and-tube heat exchanger including a large number of heat transfer tubes and fins. The plurality of utilization-side heat exchangers **52a**, **52b**, and **52c** are connected in parallel to the secondary-side switching mechanism **22**, the suction flow path **23**, and the cascade heat exchanger **35**.

(62) The second utilization pipe **56a** has one end connected to a liquid side (opposite to a gas side) of the utilization-side heat exchanger **52a** in the first utilization unit **3a**. The other end of the second utilization pipe **56a** is connected to the second connecting tube **16a**. The utilization-side expansion valve **51a** described above is provided at a halfway portion of the second utilization pipe **56a**.

(63) The utilization-side expansion valve **51a** is an electrically powered expansion valve that has an adjustable opening degree for adjusting a flow rate of the refrigerant flowing in the utilization-side heat exchanger **52a**. The utilization-side expansion valve **51a** is provided on the second utilization pipe **56a**.

(64) The first utilization pipe **57a** has one end connected to the gas side of the utilization-side heat exchanger **52a** in the first utilization unit **3a**. In the present embodiment, the first utilization pipe **57a** is connected to a portion opposite to the utilization-side expansion valve **51a** of the utilization-side heat exchanger **52a**. The first utilization pipe **57a** has the other end connected to the first connecting tube **15a**.

(65) Hereinafter, the branch circuits **14a**, **14b**, and **14c** will be described. Since the branch circuits **14b** and **14c** are configured similarly to the branch circuit **14a**, elements of the branch circuits **14b** and **14c** will not be described repeatedly, assuming that a subscript “b” or “c” will replace a subscript “a” in reference signs denoting elements of the branch circuit **14a**.

(66) The branch circuit **14a** mainly includes a junction pipe **62a**, a first branch pipe **63a**, a second branch pipe **64a**, a first regulating valve **66a**, a second regulating valve **67a**, a bypass pipe **69a**, a check valve **68a**, and a third branch pipe **61a**.

(67) The junction pipe **62a** has one end connected to the first connecting tube **15a**. The junction pipe **62a** has the other end branched to be connected with the first branch pipe **63a** and the second branch pipe **64a**.

(68) The first branch pipe **63a** has a portion opposite to the junction pipe **62** and connected to the secondary-side first connection pipe **8**. The first branch pipe **63a** is provided with the openable and closable first regulating valve **66a**.

(69) The second branch pipe **64a** has a portion opposite to the junction pipe **62** and connected to the secondary-side second connection pipe **9**. The second branch pipe **64a** is provided with the openable and closable second regulating valve **67a**.

(70) The bypass pipe **69a** is a refrigerant pipe that connects a portion of the first branch pipe **63a** closer to the secondary-side first connection pipe **8** than the first regulating valve **66a** and a portion of the second branch pipe **64a** closer to the secondary-side second connection pipe **9** than the second regulating valve **67a**. The check valve **68a** is provided at a halfway portion of the bypass pipe **69a**. The check valve **68a** allows only a refrigerant flow from the second branch pipe **64a** toward the first branch pipe **63a**, and does not allow a refrigerant flow from the first branch pipe

63a toward the second branch pipe **64a**.

(71) The third branch pipe **61a** has one end connected to the second connecting tube **16a**. The other end of the third branch pipe **61a** is connected to the secondary-side third connection pipe **7**.

(72) The first branch unit **6a** can function as follows by closing the first regulating valve **66a** and opening the second regulating valve **67a** when the cooling operation described later is performed. The first branch unit **6a** sends a refrigerant flowing into the third branch pipe **61a** through the secondary-side third connection pipe **7** to the second connecting tube **16a**. The refrigerant flowing in the second utilization pipe **56a** in the first utilization unit **3a** through the second connecting tube **16a** is sent to the utilization-side heat exchanger **52a** in the first utilization unit **3a** through the utilization-side expansion valve **51a**. The refrigerant sent to the utilization-side heat exchanger **52a** is evaporated by heat exchange with indoor air, and then flows in the first connecting tube **15a** via the first utilization pipe **57a**. The refrigerant having flowed through the first connecting tube **15a** is sent to the junction pipe **62a** of the first branch unit **6a**. The refrigerant having flowed through the junction pipe **62a** does not flow toward the first branch pipe **63a** but flows toward the second branch pipe **64a**. The refrigerant flowing in the second branch pipe **64a** passes through the second regulating valve **67a**. A part of the refrigerant that has passed through the second regulating valve **67a** is sent to the secondary-side second connection pipe **9**. The remaining part of the refrigerant that has passed through the second regulating valve **67a** flows so as to branch into the bypass pipe **69a** provided with the check valve **68a**, passes through a part of the first branch pipe **63a**, and then is sent to the secondary-side first connection pipe **8**. As a result, it is possible to increase a total flow path sectional area when the secondary-side refrigerant in a gas state evaporated in the utilization-side heat exchanger **52a** is sent to the secondary-side compressor **21**, so that a pressure loss can be reduced.

(73) When the first utilization unit **3a** cools an indoor space at the time of performing the cooling main operation and the heating main operation described later, the first branch unit **6a** can function as follows by closing the first regulating valve **66a** and opening the second regulating valve **67a**. The first branch unit **6a** sends a refrigerant flowing into the third branch pipe **61a** through the secondary-side third connection pipe **7** to the second connecting tube **16a**. The refrigerant flowing in the second utilization pipe **56a** in the first utilization unit **3a** through the second connecting tube **16a** is sent to the utilization-side heat exchanger **52a** in the first utilization unit **3a** through the utilization-side expansion valve **51a**. The refrigerant sent to the utilization-side heat exchanger **52a** is evaporated by heat exchange with indoor air, and then flows in the first connecting tube **15a** via the first utilization pipe **57a**. The refrigerant having flowed through the first connecting tube **15a** is sent to the junction pipe **62a** of the first branch unit **6a**. The refrigerant having flowed through the junction pipe **62a** flows to the second branch pipe **64a**, passes through the second regulating valve **67a**, and then is sent to the secondary-side second connection pipe **9**.

(74) The first branch unit **6a** can function as follows by closing the second regulating valve **67a** and opening the first regulating valve **66a** when the heating operation described later is performed. In the first branch unit **6a**, the refrigerant flowing into the first branch pipe **63a** through the secondary-side first connection pipe **8** passes through the first regulating valve **66a** and is sent to the junction pipe **62a**. The refrigerant having flowed through the junction pipe **62a** flows in the first utilization pipe **57a** in the utilization unit **3a** via the first connecting tube **15a** and is sent to the utilization-side heat exchanger **52a**. The refrigerant sent to the utilization-side heat exchanger **52a** radiates heat through heat exchange with indoor air, and then passes through the utilization-side expansion valve **51a** provided on the second utilization pipe **56a**. The refrigerant having passed through the second utilization pipe **56a** flows through the third branch pipe **61a** of the first branch unit **6a** via the second connecting tube **16a**, and then is sent to the secondary-side third connection pipe **7**.

(75) When the first utilization unit **3a** heats an indoor space at the time of performing the cooling main operation and the heating main operation described later, the first branch unit **6a** can function

as follows by closing the second regulating valve **67a** and opening the first regulating valve **66a**. In the first branch unit **6a**, the refrigerant flowing into the first branch pipe **63a** through the secondary-side first connection pipe **8** passes through the first regulating valve **66a** and is sent to the junction pipe **62a**. The refrigerant having flowed through the junction pipe **62a** flows in the first utilization pipe **57a** in the utilization unit **3a** via the first connecting tube **15a** and is sent to the utilization-side heat exchanger **52a**. The refrigerant sent to the utilization-side heat exchanger **52a** radiates heat through heat exchange with indoor air, and then passes through the utilization-side expansion valve **51a** provided on the second utilization pipe **56a**. The refrigerant having passed through the second utilization pipe **56a** flows through the third branch pipe **61a** of the first branch unit **6a** via the second connecting tube **16a**, and then is sent to the secondary-side third connection pipe **7**.

(76) The first branch unit **6a**, as well as the second branch unit **6b** and the third branch unit **6c**, similarly have such a function. Accordingly, the first branch unit **6a**, the second branch unit **6b**, and the third branch unit **6c** can individually switchably cause the utilization-side heat exchangers **52a**, **52b**, and **52c** to function as a refrigerant evaporator or a refrigerant radiator.

(4) Primary-Side Unit

(77) The primary-side unit **5** is disposed in a space different from a space provided with the utilization units **3a**, **3b**, and **3c** and the branch units **6a**, **6b**, and **6c**, on a roof, or the like.

(78) The primary-side unit **5** includes a part of the primary-side refrigerant circuit **5a** described above, a primary-side fan **75**, various sensors, and a primary-side control unit **70**, and a primary-side casing (not illustrated).

(79) The primary-side unit **5** includes, as a part of the primary-side refrigerant circuit **5a**, the primary-side compressor **71**, the primary-side switching mechanism **72**, the primary-side heat exchanger **74**, the primary-side first expansion valve **76**, the primary-side subcooling heat exchanger **103**, the primary-side subcooling circuit **104**, the primary-side subcooling expansion valve **104a**, the first liquid shutoff valve **108**, the first gas shutoff valve **109**, and the primary-side accumulator **105** in the primary-side casing.

(80) The primary-side fan **75** is provided in the primary-side unit **5**, and generates an air flow of guiding outdoor air into the primary-side heat exchanger **74**, and exhausting, to outdoors, air obtained after heat exchange with the primary-side refrigerant flowing in the primary-side heat exchanger **74**. The primary-side fan **75** is driven by a primary-side fan motor **75a**.

(81) The primary-side unit **5** is provided with the various sensors. Specifically, there are provided an outdoor air temperature sensor **77** that detects a temperature of outdoor air before passing through the primary-side heat exchanger **74**, a primary-side discharge pressure sensor **78** that detects a pressure of the primary-side refrigerant discharged from the primary-side compressor **71**, a primary-side suction pressure sensor **79** that detects a pressure of the primary-side refrigerant sucked into the primary-side compressor **71**, a primary-side suction temperature sensor **81** that detects a temperature of the primary-side refrigerant sucked into the primary-side compressor **71**, and a primary-side heat exchange temperature sensor **82** that detects a temperature of the refrigerant flowing in the primary-side heat exchanger **74**.

(82) The primary-side control unit **70** controls motion of the elements **71** (**71a**), **72**, **75** (**75a**), **76**, and **104a** provided in the primary-side unit **5**. The primary-side control unit **70** includes a processor such as a CPU or a microcomputer provided to control the primary-side unit **5** and a memory, so as to transmit and receive control signals and the like to and from a remote controller (not illustrated), and to transmit and receive control signals and the like between a cascade-side control unit **20** in a cascade unit **2**, branch unit control units **60a**, **60b**, and **60c**, and utilization-side control units **50a**, **50b**, and **50c**.

(5) Cascade Unit

(83) The cascade unit **2** is disposed in a space different from a space provided with the utilization units **3a**, **3b**, and **3c** and the branch units **6a**, **6b**, and **6c**, on a roof, or the like.

(84) The cascade unit **2** is connected to the branch units **6a**, **6b**, and **6c** via the connection pipes **7**, **8**, and **9**, to constitute a part of the secondary-side refrigerant circuit **10**. The cascade unit **2** is connected to the primary-side unit **5** via the primary-side first connection pipe **111** and the primary-side second connection pipe **112**, to constitute a part of the primary-side refrigerant circuit **5a**.

(85) The cascade unit **2** mainly includes the cascade circuit **12** described above, various sensors, the cascade-side control unit **20**, the second liquid shutoff valve **106**, the second refrigerant pipe **114**, the primary-side second expansion valve **102**, the first refrigerant pipe **113**, and the second gas shutoff valve **107** that constitute a part of the primary-side refrigerant circuit **5a**, the cascade casing (not illustrated), and the like.

(86) The cascade unit **2** is provided with a secondary-side suction pressure sensor **37** that detects a pressure of the secondary-side refrigerant on the suction side of the secondary-side compressor **21**, a secondary-side discharge pressure sensor **38** that detects a pressure of the secondary-side refrigerant on the discharge side of the secondary-side compressor **21**, a secondary-side discharge temperature sensor **39** that detects a temperature of the secondary-side refrigerant on the discharge side of the secondary-side compressor **21**, a secondary-side suction temperature sensor **88** that detects a temperature of the secondary-side refrigerant on the suction side of the secondary-side compressor **21**, a secondary-side cascade temperature sensor **83** that detects a temperature of the secondary-side refrigerant flowing between the secondary-side flow path **35a** of the cascade heat exchanger **35** and the cascade expansion valve **36**, a receiver outlet temperature sensor **84** that detects a temperature of the secondary-side refrigerant flowing between the secondary-side receiver **45** and the secondary-side subcooling heat exchanger **47**, a bypass circuit temperature sensor **85** that detects a temperature of the secondary-side refrigerant flowing downstream of the bypass expansion valve **46a** in the bypass circuit **46**, a subcooling outlet temperature sensor **86** that detects a temperature of the secondary-side refrigerant flowing between the secondary-side subcooling heat exchanger **47** and the third shutoff valve **31**, and a subcooling circuit temperature sensor **87** that detects a temperature of the secondary-side refrigerant flowing at an outlet of the secondary-side subcooling heat exchanger **47** in the secondary-side subcooling circuit **48**.

(87) The cascade-side control unit **20** controls motion of the elements **21** (**21a**), **22**, **36**, **44**, **46a**, **48a**, and **102** provided in the cascade casing of the cascade unit **2**. The cascade-side control unit **20** includes a processor such as a CPU or a microcomputer provided to control the cascade unit **2** and a memory, so as to transmit and receive control signals and the like between the primary-side control unit **70** in the primary-side unit **5**, the utilization-side control units **50a**, **50b**, and **50c** in the utilization units **3a**, **3b**, and **3c**, and the branch unit control units **60a**, **60b**, and **60c**.

(88) In such a manner, the cascade-side control unit **20** can control not only the elements constituting the cascade circuit **12** of the secondary-side refrigerant circuit **10** but also the primary-side second expansion valve **102** constituting a part of the primary-side refrigerant circuit **5a**. Therefore, the cascade-side control unit **20** controls a valve opening degree of the primary-side second expansion valve **102** on the basis of a condition of the cascade circuit **12** controlled by the cascade-side control unit **20**, so as to bring the condition of the cascade circuit **12** closer to a desired condition. Specifically, it is possible to control an amount of heat received by the secondary-side refrigerant flowing in the secondary-side flow path **35a** of the cascade heat exchanger **35** in the cascade circuit **12** from the primary-side refrigerant flowing in the primary-side flow path **35b** of the cascade heat exchanger **35** or an amount of heat given by the secondary-side refrigerant to the primary-side refrigerant.

(6) Utilization Unit

(89) The utilization units **3a**, **3b**, and **3c** are installed by being embedded in or being suspended from a ceiling on an indoor space of a building or the like, or by being hung on a wall surface in the indoor space, or the like.

(90) The utilization units **3a**, **3b**, and **3c** are connected to the cascade unit **2** via the connection pipes **7**, **8**, and **9**.

(91) The utilization units **3a**, **3b**, and **3c** respectively include the utilization circuits **13a**, **13b**, and **13c** constituting a part of the secondary-side refrigerant circuit **10**.

(92) Hereinafter, the utilization units **3a**, **3b**, and **3c** will be described in terms of their configurations. The second utilization unit **3b** and the third utilization unit **3c** are configured similarly to the first utilization unit **3a**. The configuration of only the first utilization unit **3a** will thus be described here. As for the configuration of each of the second utilization unit **3b** and the third utilization unit **3c**, elements will be denoted by reference signs obtained by replacing a subscript “a” in reference signs of elements of the first utilization unit **3a** with a subscript “b” or “c”, and these elements will not be described repeatedly.

(93) The first utilization unit **3a** mainly includes the utilization circuit **13a** described above, an indoor fan **53a**, the utilization-side control unit **50a**, and various sensors. Note that the indoor fan **53a** includes an indoor fan motor **54a**.

(94) The indoor fan **53a** generates an air flow by sucking indoor air into the unit and supplying the indoor space with supply air obtained after heat exchange with the refrigerant flowing in the utilization-side heat exchanger **52a**. The indoor fan **53a** is driven by the indoor fan motor **54a**.

(95) The utilization unit **3a** is provided with a liquid-side temperature sensor **58a** that detects a temperature of a refrigerant on the liquid side of the utilization-side heat exchanger **52a**. The utilization unit **3a** is further provided with an indoor temperature sensor **55a** that detects an indoor temperature as temperature of air introduced from the indoor space before passing through the utilization-side heat exchanger **52a**.

(96) The utilization-side control unit **50a** controls motion of the elements **51a** and **53a** (**54a**) constituting the utilization unit **3a**. The utilization-side control unit **50a** includes a processor such as a CPU or a microcomputer provided to control the utilization unit **3a** and a memory, so as to transmit and receive control signals and the like to and from the remote controller (not illustrated), and to transmit and receive control signals and the like between the cascade-side control unit **20** in the cascade unit **2**, the branch unit control units **60a**, **60b**, and **60c**, and the primary-side control unit **70** in the primary-side unit **5**.

(97) Note that the second utilization unit **3b** includes the utilization circuit **13b**, an indoor fan **53b**, the utilization-side control unit **50b**, and an indoor fan motor **54b**. The third utilization unit **3c** includes the utilization circuit **13c**, an indoor fan **53c**, the utilization-side control unit **50c**, and an indoor fan motor **54c**.

(7) Branch Unit

(98) The branch units **6a**, **6b**, and **6c** are installed in a space above a ceiling of an indoor space of a building or the like.

(99) Each of the branch units **6a**, **6b**, and **6c** is connected to a corresponding one of the utilization units **3a**, **3b**, and **3c** one by one. The branch units **6a**, **6b**, and **6c** are connected to the cascade unit **2** via the connection pipes **7**, **8**, and **9**.

(100) Next, the branch units **6a**, **6b**, and **6c** will be described next in terms of their configurations. The second branch unit **6b** and the third branch unit **6c** are configured similarly to the first branch unit **6a**. The configuration of only the first branch unit **6a** will thus be described here. As for the configuration of each of the second branch unit **6b** and the third branch unit **6c**, elements will be denoted by reference signs obtained by replacing a subscript “a” in reference signs of elements of the first branch unit **6a** with a subscript “b” or “c”, and these elements will not be described repeatedly.

(101) The first branch unit **6a** mainly includes the branch circuit **14a** described above and the branch unit control unit **60a**.

(102) The branch unit control unit **60a** controls motion of the elements **66a** and **67a** constituting the branch unit **6a**. The branch unit control unit **60a** includes a processor such as a CPU or a microcomputer provided to control the branch unit **6a** and a memory, so as to transmit and receive control signals and the like to and from the remote controller (not depicted), and to transmit and

receive control signals and the like between the cascade-side control unit **20** in the cascade unit **2**, the utilization units **3a**, **3b**, and **3c**, and the primary-side control unit **70** in the primary-side unit **5**. (103) Note that the second branch unit **6b** includes the branch circuit **14b** and the branch unit control unit **60b**. The third branch unit **6c** includes the branch circuit **14c** and the branch unit control unit **60c**.

(8) Control Unit

(104) In the refrigeration cycle apparatus **1**, the heat cascade-side control unit **20**, the utilization-side control units **50a**, **50b**, and **50c**, the branch unit control units **60a**, **60b**, and **60c**, and the primary-side control unit **70** described above are communicably connected to each other in a wired or wireless manner to constitute a control unit **80**. Therefore, the control unit **80** controls motion of the elements **21**(**21a**), **22**, **36**, **44**, **46a**, **48a**, **51a**, **51b**, **51c**, **53a**, **53b**, **53c** (**54a**, **54b**, **54c**), **66a**, **66b**, **66c**, **67a**, **67b**, **67c**, **71** (**71a**), **72**, **75** (**75a**), **76**, **104a** on the basis of detection information of various sensors **37**, **38**, **39**, **83**, **84**, **85**, **86**, **87**, **88**, **77**, **78**, **79**, **81**, **82**, **58a**, **58b**, **58c**, and the like, and instruction information received from a remote controller (not illustrated) and the like.

(9) Motion of Refrigeration Cycle Apparatus

(105) Next, motion of the refrigeration cycle apparatus **1** will be described with reference to FIGS. **3** to **6**.

(106) The refrigeration cycle operation of the refrigeration cycle apparatus **1** can be mainly divided into cooling operation, heating operation, cooling main operation, and heating main operation.

(107) Here, the cooling operation is refrigeration cycle operation in which only the utilization unit in which the utilization-side heat exchanger functions as a refrigerant evaporator exists, and the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant with respect to an evaporation load of the entire utilization unit.

(108) Here, the heating operation is refrigeration cycle operation in which only the utilization unit in which the utilization-side heat exchanger functions as a refrigerant radiator exists, and the cascade heat exchanger **35** functions as an evaporator for the secondary-side refrigerant with respect to a radiation load of the entire utilization unit.

(109) The cooling main operation is operation in which the utilization unit in which the utilization-side heat exchanger functions as a refrigerant evaporator and the utilization unit in which the utilization-side heat exchanger functions as a refrigerant radiator are mixed. The cooling main operation is refrigeration cycle operation in which, when an evaporation load is a main heat load of the entire utilization unit, the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant in order to process the evaporation load of the entire utilization unit.

(110) The heating main operation is operation in which the utilization unit in which the utilization-side heat exchanger functions as a refrigerant evaporator and the utilization unit in which the utilization-side heat exchanger functions as a refrigerant radiator are mixed. The heating main operation is refrigeration cycle operation in which, when a radiation load is a main heat load of the entire utilization unit, the cascade heat exchanger **35** functions as an evaporator for the secondary-side refrigerant in order to process the radiation load of the entire utilization unit.

(111) Note that the motion of the refrigeration cycle apparatus **1** including the refrigeration cycle operation is performed by the control unit **80** described above.

(9-1) Cooling Operation

(112) During the cooling operation, for example, all of the utilization-side heat exchangers **52a**, **52b**, and **52c** in the utilization units **3a**, **3b**, and **3c** functions as a refrigerant evaporator, and the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant. In the cooling operation, the primary-side refrigerant circuit **5a** and the secondary-side refrigerant circuit **10** of the refrigeration cycle apparatus **1** are configured as illustrated in FIG. **3**. Note that arrows attached to the primary-side refrigerant circuit **5a** and arrows attached to the secondary-side refrigerant circuit **10** in FIG. **3** indicate flows of the refrigerant during the cooling operation.

(113) Specifically, in the primary-side unit **5**, the primary-side switching mechanism **72** is switched

to the fifth connection state to cause the cascade heat exchanger **35** to function as an evaporator for the primary-side refrigerant. The fifth connection state of the primary-side switching mechanism **72** is depicted by solid lines in the primary-side switching mechanism **72** in FIG. **3**. Accordingly, in the primary-side unit **5**, the primary-side refrigerant discharged from the primary-side compressor **71** passes through the primary-side switching mechanism **72** and exchanges heat with outdoor air supplied from the primary-side fan **75** in the primary-side heat exchanger **74** to be condensed. The primary-side refrigerant condensed in the primary-side heat exchanger **74** passes through the primary-side first expansion valve **76** controlled into a fully opened state, and a part of the refrigerant flows toward the first liquid shutoff valve **108** through the primary-side subcooling heat exchanger **103**, and another part of the refrigerant branches into the primary-side subcooling circuit **104**. The refrigerant flowing in the primary-side subcooling circuit **104** is decompressed when passing through the primary-side subcooling expansion valve **104a**. The refrigerant flowing from the primary-side first expansion valve **76** toward the first liquid shutoff valve **108** exchanges heat with the refrigerant decompressed by the primary-side subcooling expansion valve **104a** and flowing in the primary-side subcooling circuit **104** in the primary-side subcooling heat exchanger **103**, and is cooled until reaching a subcooled state. The refrigerant in the subcooled state flows through the primary-side first connection pipe **111**, the second liquid shutoff valve **106**, and the second refrigerant pipe **114** in that order, and is decompressed when passing through the primary-side second expansion valve **102**. Here, a valve opening degree of the primary-side second expansion valve **102** is controlled such that a degree of superheating of the primary-side refrigerant sucked into the primary-side compressor **71** satisfies a predetermined condition. When flowing in the primary-side flow path **35b** of the cascade heat exchanger **35**, the primary-side refrigerant decompressed by the primary-side second expansion valve **102** evaporates by exchanging heat with the secondary-side refrigerant in through the secondary-side flow path **35a**, and flows toward the second gas shutoff valve **107** through the first refrigerant pipe **113**. The refrigerant having passed through the second gas shutoff valve **107** passes through the primary-side second connection pipe **112** and the first gas shutoff valve **109**, and then reaches the primary-side switching mechanism **72**. The refrigerant having passed through the primary-side switching mechanism **72** joins the refrigerant having flowed through the primary-side subcooling circuit **104**, and is then sucked into the primary-side compressor **71** via the primary-side accumulator **105**.

(114) In the cascade unit **2**, by switching the secondary-side switching mechanism **22** to the first connection state, the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant. In the first connection state of the secondary-side switching mechanism **22**, the discharge flow path **24** and the third pipe **25** are connected by the first switching valve **22a**, and the first pipe **28** and the suction flow path **23** are connected by the second switching valve **22b**. In the first to third utilization units **3a**, **3b**, **3c**, the second regulating valves **67a**, **67b**, **67c** are controlled to the opened state. Accordingly, all of the utilization-side heat exchangers **52a**, **52b**, and **52c** in the utilization units **3a**, **3b**, and **3c** function as refrigerant evaporators. All of the utilization-side heat exchangers **52a**, **52b**, and **52c** of the utilization units **3a**, **3b**, and **3c** and the suction side of the secondary-side compressor **21** of the cascade unit **2** are connected via the first utilization pipes **57a**, **57b**, and **57c**, the first connecting tubes **15a**, **15b**, and **15c**, the junction pipes **62a**, **62b**, and **62c**, the second branch pipes **64a**, **64b**, and **64c**, the bypass pipes **69a**, **69b**, and **69c**, a part of the first branch pipes **63a**, **63b**, and **63c**, the secondary-side first connection pipe **8**, and the secondary-side second connection pipe **9**. In addition, an opening degree of the secondary-side subcooling expansion valve **48a** is controlled such that a degree of subcooling of the secondary-side refrigerant flowing through the outlet of the secondary-side subcooling heat exchanger **47** toward the secondary-side third connection pipe **7** satisfies a predetermined condition. The bypass expansion valve **46a** is controlled to the closed state. In the utilization units **3a**, **3b**, and **3c**, the opening degrees of the utilization-side expansion valves **51a**, **51b**, and **51c** are adjusted.

(115) In the cooling operation, the secondary-side refrigerant circuit **10** controls capacity, for

example, by controlling a frequency of the secondary-side compressor **21** so that evaporation temperature of the secondary-side refrigerant in the utilization-side heat exchangers **52a**, **52b**, and **52c** becomes a predetermined secondary-side evaporation target temperature. The opening degree of the cascade expansion valve **36** is adjusted such that the secondary-side refrigerant flowing in the cascade heat exchanger **35** has a critical pressure or less. The primary-side refrigerant circuit **5a** controls capacity, for example, by controlling a frequency of the primary-side compressor **71** such that evaporation temperature of the primary-side refrigerant in the primary-side flow path **35b** of the cascade heat exchanger **35** becomes a predetermined primary-side evaporation target temperature. In such a manner, in the cooling operation, either or both of the control for increasing the valve opening degree of the cascade expansion valve **36** and the control for increasing the frequency of the primary-side compressor **71** in the primary-side refrigerant circuit **5a** are executed, and thus, the carbon dioxide refrigerant flowing in the cascade heat exchanger **35** is controlled so as not to exceed a critical point.

(116) In such a secondary-side refrigerant circuit **10**, a secondary-side high-pressure refrigerant compressed and discharged by the secondary-side compressor **21** is sent to the secondary-side flow path **35a** of the cascade heat exchanger **35** through the first switching valve **22a** of the secondary-side switching mechanism **22**. The secondary-side high-pressure refrigerant flowing in the secondary-side flow path **35a** of the cascade heat exchanger **35** radiates heat, and the primary-side refrigerant flowing in the primary-side flow path **35b** of the cascade heat exchanger **35** is evaporated. The secondary-side refrigerant having radiated heat in the cascade heat exchanger **35** passes through the cascade expansion valve **36** whose opening degree is adjusted, and then flows into the secondary-side receiver **45**. A part of the refrigerant flowing out of the secondary-side receiver **45** branches and flows into the secondary-side subcooling circuit **48**, is decompressed in the secondary-side subcooling expansion valve **48a**, and then joins the suction flow path **23**. In the secondary-side subcooling heat exchanger **47**, another part of the refrigerant having flowed out of the secondary-side receiver **45** is cooled by the refrigerant flowing in the secondary-side subcooling circuit **48**, and is then sent to the secondary-side third connection pipe **7** through the third shutoff valve **31**.

(117) Then, the refrigerant sent to the secondary-side third connection pipe **7** is branched into three portions to pass through the third branch pipes **61a**, **61b**, and **61c** of the first to third branch units **6a**, **6b**, and **6c**. Thereafter, the refrigerant having flowed through the second connecting tubes **16a**, **16b**, and **16c** is sent to the second utilization pipes **56a**, **56b**, and **56c** of the first to third utilization units **3a**, **3b**, and **3c**. The refrigerant sent to the second utilization pipes **56a**, **56b**, and **56c** is sent to the utilization-side expansion valves **51a**, **51b**, and **51c** in the utilization units **3a**, **3b**, and **3c**.

(118) Then, the refrigerant having passed through the utilization-side expansion valves **51a**, **51b**, and **51c** whose opening degrees are adjusted exchanges heat with indoor air supplied by the indoor fans **53a**, **53b**, and **53c** in the utilization-side heat exchangers **52a**, **52b**, and **52c**. The refrigerant flowing in the utilization-side heat exchangers **52a**, **52b**, and **52c** is thus evaporated into a low-pressure gas refrigerant. The indoor air is cooled and is supplied into the indoor space. The indoor space is thus cooled. The low-pressure gas refrigerant evaporated in the utilization-side heat exchangers **52a**, **52b**, and **52c** flows in the first utilization pipes **57a**, **57b**, and **57c**, flows through the first connecting tubes **15a**, **15b**, and **15c**, and then is sent to the junction pipes **62a**, **62b**, and **62c** of the first to third branch units **6a**, **6b**, and **6c**.

(119) Then, the low-pressure gas refrigerant sent to the junction pipes **62a**, **62b**, and **62c** flows to the second branch pipes **64a**, **64b**, and **64c**. A part of the refrigerant that has passed through the second regulating valves **67a**, **67b**, and **67c** in the second branch pipes **64a**, **64b**, and **64c** is sent to the secondary-side second connection pipe **9**. The remaining part of the refrigerant that has passed through the second regulating valves **67a**, **67b**, and **67c** passes through the bypass pipes **69a**, **69b**, and **69c**, flows through a part of the first branch pipes **63a**, **63b**, and **63c**, and then is sent to the secondary-side first connection pipe **8**.

(120) Then, the low-pressure gas refrigerant sent to the secondary-side first connection pipe **8** and the secondary-side second connection pipe **9** is returned to the suction side of the secondary-side compressor **21** through the first shutoff valve **32**, the second shutoff valve **33**, the first pipe **28**, the second pipe **29**, the second switching valve **22b** of the secondary-side switching mechanism **22**, the suction flow path **23**, and the secondary-side accumulator **30**.

(121) Motion during the cooling operation is performed in such a manner.

(9-2) Heating Operation

(122) During the heating operation, each of the utilization-side heat exchangers **52a**, **52b**, and **52c** in the utilization units **3a**, **3b**, and **3c** functions as a refrigerant radiator. In the heating operation, the cascade heat exchanger **35** operates to function as an evaporator for the secondary-side refrigerant. In the heating operation, the primary-side refrigerant circuit **5a** and the secondary-side refrigerant circuit **10** of the refrigeration cycle apparatus **1** are configured as illustrated in FIG. **4**. Arrows attached to the primary-side refrigerant circuit **5a** and arrows attached to the secondary-side refrigerant circuit **10** in FIG. **4** indicate flows of the refrigerant during the heating operation.

(123) Specifically, in the primary-side unit **5**, the primary-side switching mechanism **72** is switched to a sixth operating state to cause the cascade heat exchanger **35** to function as a radiator for the primary-side refrigerant. The sixth operating state of the primary-side switching mechanism **72** is a connection state depicted by broken lines in the primary-side switching mechanism **72** in FIG. **4**. Accordingly, in the primary-side unit **5**, the primary-side refrigerant discharged from the primary-side compressor **71**, having passed through the primary-side switching mechanism **72** and the first gas shutoff valve **109** passes through the primary-side second connection pipe **112** and the second gas shutoff valve **107** and is sent to the primary-side flow path **35b** of the cascade heat exchanger **35**. The refrigerant flowing in the primary-side flow path **35b** of the cascade heat exchanger **35** is condensed by exchanging heat with the secondary-side refrigerant flowing in the secondary-side flow path **35a**. When flowing in the second refrigerant pipe **114**, the primary-side refrigerant condensed in the cascade heat exchanger **35** passes through the primary-side second expansion valve **102** controlled to the fully opened state. The refrigerant that has passed through the primary-side second expansion valve **102** flows through the second liquid shutoff valve **106**, the primary-side first connection pipe **111**, the first liquid shutoff valve **108**, and the primary-side subcooling heat exchanger **103** in that order, and is decompressed by the primary-side first expansion valve **76**. During heating operation, the primary-side subcooling expansion valve **104a** is controlled to the closed state. Accordingly, the refrigerant does not flow to the primary-side subcooling circuit **104** and does not exchange heat in the primary-side subcooling heat exchanger **103**. The valve opening degree of the primary-side first expansion valve **76** is controlled such that, for example, the degree of superheating of the refrigerant sucked into the primary-side compressor **71** satisfies a predetermined condition. The refrigerant decompressed by the primary-side first expansion valve **76** evaporates by exchanging heat with outdoor air supplied from the primary-side fan **75** in the primary-side heat exchanger **74**, passes through the primary-side switching mechanism **72** and the primary-side accumulator **105**, and is sucked into the primary-side compressor **71**.

(124) In the cascade unit **2**, the secondary-side switching mechanism **22** is switched to the second connection state. The cascade heat exchanger **35** thus functions as an evaporator for the secondary-side refrigerant. In the second connection state of the secondary-side switching mechanism **22**, the discharge flow path **24** and the first pipe **28** are connected by the second switching valve **22b**, and the third pipe **25** and the suction flow path **23** are connected by the first switching valve **22a**. The opening degree of the cascade expansion valve **36** is adjusted. In the first to third branch units **6a**, **6b**, and **6c**, the first regulating valves **66a**, **66b**, and **66c** are controlled to the opened state, and the second regulating valves **67a**, **67b**, and **67c** are controlled to the closed state. Accordingly, all of the utilization-side heat exchangers **52a**, **52b**, and **52c** in the utilization units **3a**, **3b**, and **3c** function as refrigerant radiators. The utilization-side heat exchangers **52a**, **52b**, and **52c** in the utilization units **3a**, **3b**, and **3c** and the discharge side of the secondary-side compressor **21** in the cascade unit **2** are

connected via the discharge flow path **24**, the first pipe **28**, the secondary-side first connection pipe **8**, the first branch pipes **63a**, **63b**, and **63c**, the junction pipes **62a**, **62b**, and **62c**, the first connecting tubes **15a**, **15b**, and **15c**, and the first utilization pipes **57a**, **57b**, and **57c**. The secondary-side subcooling expansion valve **48a** and the bypass expansion valve **46a** are controlled to the closed state. In the utilization units **3a**, **3b**, and **3c**, the opening degrees of the utilization-side expansion valves **51a**, **51b**, and **51c** are adjusted.

(125) During the heating operation, the secondary-side refrigerant circuit **10** controls capacity on the secondary-side compressor **21** so as to achieve a frequency at which the loads in the utilization-side heat exchangers **52a**, **52b**, and **52c** can be processed. As a result, in the heating operation, the secondary-side refrigerant discharged from the secondary-side compressor **21** is controlled to be in a critical state exceeding the critical pressure. The primary-side refrigerant circuit **5a** controls capacity, for example, by controlling the frequency of the primary-side compressor **71** such that condensation temperature of the primary-side refrigerant in the primary-side flow path **35b** of the cascade heat exchanger **35** becomes a predetermined primary-side condensation target temperature.

(126) In such a secondary-side refrigerant circuit **10**, the high-pressure refrigerant compressed and discharged by the secondary-side compressor **21** is sent to the first pipe **28** through the second switching valve **22b** of the secondary-side switching mechanism **22**. The refrigerant sent to the first pipe **28** is sent to the secondary-side first connection pipe **8** through the first shutoff valve **32**.

(127) Then, the high-pressure refrigerant sent to the secondary-side first connection pipe **8** is branched into three portions to be sent to the first branch pipes **63a**, **63b**, and **63c** in the utilization units **3a**, **3b**, and **3c** which are utilization units in operation. The high-pressure refrigerant sent to the first branch pipes **63a**, **63b**, and **63c** passes through the first regulating valves **66a**, **66b**, and **66c**, and flows in the junction pipes **62a**, **62b**, and **62c**. Thereafter, the refrigerant having flowed in the first connecting tubes **15a**, **15b**, and **15c** and the first utilization pipes **57a**, **57b**, and **57c** is sent to the utilization-side heat exchangers **52a**, **52b**, and **52c**.

(128) Then, the high-pressure refrigerant sent to the utilization-side heat exchangers **52a**, **52b**, and **52c** exchanges heat with indoor air supplied by the indoor fans **53a**, **53b**, and **53c** in the utilization-side heat exchangers **52a**, **52b**, and **52c**. The refrigerant flowing in the utilization-side heat exchangers **52a**, **52b**, and **52c** thus radiates heat. The indoor air is heated and supplied into the indoor space. The indoor space is thus heated. The refrigerant having radiated heat in the utilization-side heat exchangers **52a**, **52b**, and **52c** flows in the second utilization pipes **56a**, **56b**, and **56c** and passes through the utilization-side expansion valves **51a**, **51b**, and **51c** whose opening degrees are adjusted. The secondary-side refrigerant that has passed through the utilization-side expansion valves **51a**, **51b**, and **51c** has the critical pressure or less. Thereafter, the refrigerant having flowed through the second connecting tubes **16a**, **16b**, and **16c** flows in the third branch pipes **61a**, **61b**, and **61c** of the branch units **6a**, **6b**, and **6c**.

(129) The refrigerant sent to the third branch pipes **61a**, **61b**, and **61c** is sent to the secondary-side third connection pipe **7** to join.

(130) The refrigerant sent to the secondary-side third connection pipe **7** passes through the third shutoff valve **31** and then is sent to the cascade expansion valve **36**. The flow rate of the refrigerant sent to the cascade expansion valve **36** is adjusted at the cascade expansion valve **36**, and then, the refrigerant is sent to the cascade heat exchanger **35**. In the cascade heat exchanger **35**, the secondary-side refrigerant flowing in the secondary-side flow path **35a** is evaporated into a low-pressure gas refrigerant and is sent to the secondary-side switching mechanism **22**, and the primary-side refrigerant flowing in the primary-side flow path **35b** of the cascade heat exchanger **35** is condensed. Then, the secondary-side low-pressure gas refrigerant sent to the first switching valve **22a** of the secondary-side switching mechanism **22** is returned to the suction side of the secondary-side compressor **21** through the suction flow path **23** and the secondary-side accumulator **30**.

(131) Motion during the heating operation is performed in such a manner.

(9-3) Cooling Main Operation

(132) During the cooling main operation, for example, the utilization-side heat exchangers **52a** and **52b** in the utilization units **3a** and **3b** function as refrigerant evaporators, and the utilization-side heat exchanger **52c** in the utilization unit **3c** functions as a refrigerant radiator. In the cooling main operation, the cascade heat exchanger **35** functions as a radiator for the secondary-side refrigerant. In the cooling main operation, the primary-side refrigerant circuit **5a** and the secondary-side refrigerant circuit **10** of the refrigeration cycle apparatus **1** are configured as illustrated in FIG. 5. Arrows attached to the primary-side refrigerant circuit **5a** and arrows attached to the secondary-side refrigerant circuit **10** in FIG. 5 indicate flows of the refrigerant during the cooling main operation.

(133) Specifically, in the primary-side unit **5**, the primary-side switching mechanism **72** is switched to the fifth connection state (the state depicted by solid lines in the primary-side switching mechanism **72** in FIG. 5) to cause the cascade heat exchanger **35** to function as an evaporator for the primary-side refrigerant. Accordingly, in the primary-side unit **5**, the primary-side refrigerant discharged from the primary-side compressor **71** passes through the primary-side switching mechanism **72** and exchanges heat with outdoor air supplied from the primary-side fan **75** in the primary-side heat exchanger **74** to be condensed. The primary-side refrigerant condensed in the primary-side heat exchanger **74** passes through the primary-side first expansion valve **76** controlled into a fully opened state, and a part of the refrigerant flows toward the first liquid shutoff valve **108** through the primary-side subcooling heat exchanger **103**, and another part of the refrigerant branches into the primary-side subcooling circuit **104**. The refrigerant flowing in the primary-side subcooling circuit **104** is decompressed when passing through the primary-side subcooling expansion valve **104a**. The refrigerant flowing from the primary-side first expansion valve **76** toward the first liquid shutoff valve **108** exchanges heat with the refrigerant decompressed by the primary-side subcooling expansion valve **104a** and flowing in the primary-side subcooling circuit **104** in the primary-side subcooling heat exchanger **103**, and is cooled until reaching a subcooled state. The refrigerant in the subcooled state flows through the primary-side first connection pipe **111**, the second liquid shutoff valve **106**, and the second refrigerant pipe **114** in that order, and is decompressed by the primary-side second expansion valve **102**. At this time, for example, a valve opening degree of the primary-side second expansion valve **102** is controlled such that the degree of superheating of the refrigerant sucked into the primary-side compressor **71** satisfies a predetermined condition. When flowing in the primary-side flow path **35b** of the cascade heat exchanger **35**, the primary-side refrigerant decompressed by the primary-side second expansion valve **102** evaporates by exchanging heat with the secondary-side refrigerant in through the secondary-side flow path **35a**, and flows toward the second gas shutoff valve **107** through the first refrigerant pipe **113**. The refrigerant having passed through the second gas shutoff valve **107** passes through the primary-side second connection pipe **112** and the first gas shutoff valve **109**, and then reaches the primary-side switching mechanism **72**. The refrigerant having passed through the primary-side switching mechanism **72** joins the refrigerant having flowed through the primary-side subcooling circuit **104**, and is then sucked into the primary-side compressor **71** via the primary-side accumulator **105**.

(134) In the cascade unit **2**, the secondary-side switching mechanism **22** is switched to the third connection state in which the discharge flow path **24** and the third pipe **25** are connected by the first switching valve **22a** and the discharge flow path **24** and the first pipe **28** are connected by the second switching valve **22b** to cause the cascade heat exchanger **35** to function as a radiator for the secondary-side refrigerant. The opening degree of the cascade expansion valve **36** is adjusted. In the first to third branch units **6a**, **6b**, and **6c**, the first regulating valve **66c** and the second regulating valves **67a** and **67b** are controlled to the opened state, and the first regulating valves **66a** and **66b** and the second regulating valve **67c** are controlled to the closed state. Accordingly, the utilization-side heat exchangers **52a** and **52b** in the utilization units **3a** and **3b** function as refrigerant

evaporators, and the utilization-side heat exchanger **52c** in the utilization unit **3c** functions as a refrigerant radiator. The utilization-side heat exchangers **52a** and **52b** in the utilization units **3a** and **3b** and the suction side of the secondary-side compressor **21** in the cascade unit **2** are connected via the secondary-side second connection pipe **9**, and the utilization-side heat exchanger **52c** in the utilization unit **3c** and the discharge side of the secondary-side compressor **21** in the cascade unit **2** are connected via the secondary-side first connection pipe **8**. In addition, an opening degree of the secondary-side subcooling expansion valve **48a** is controlled such that a degree of subcooling of the secondary-side refrigerant flowing through the outlet of the secondary-side subcooling heat exchanger **47** toward the secondary-side third connection pipe **7** satisfies a predetermined condition. The bypass expansion valve **46a** is controlled to the closed state. In the utilization units **3a**, **3b**, and **3c**, the opening degrees of the utilization-side expansion valves **51a**, **51b**, and **51c** are adjusted.

(135) In the cooling main operation, the secondary-side refrigerant circuit **10** controls capacity, for example, by controlling the frequency of the secondary-side compressor **21** such that evaporation temperature in a heat exchanger functioning as an evaporator for the secondary-side refrigerant between the utilization-side heat exchanger **52a**, **52b**, and **52c** becomes a predetermined secondary-side evaporation target temperature. The opening degree of the cascade expansion valve **36** is adjusted such that the secondary-side refrigerant flowing in the cascade heat exchanger **35** has a critical pressure or less. The primary-side refrigerant circuit **5a** controls capacity, for example, by controlling a frequency of the primary-side compressor **71** such that evaporation temperature of the primary-side refrigerant in the primary-side flow path **35b** of the cascade heat exchanger **35** becomes a predetermined primary-side evaporation target temperature. In such a manner, in the cooling operation, either or both of the control for increasing the valve opening degree of the cascade expansion valve **36** and the control for increasing the frequency of the primary-side compressor **71** in the primary-side refrigerant circuit **5a** are executed, and thus, the carbon dioxide refrigerant flowing in the cascade heat exchanger **35** is controlled so as not to exceed a critical point.

(136) In such a secondary-side refrigerant circuit **10**, a part of the secondary-side high-pressure refrigerant compressed and discharged by the secondary-side compressor **21** is sent to the secondary-side first connection pipe **8** through the second switching valve **22b** of the secondary-side switching mechanism **22**, the first pipe **28**, and the first shutoff valve **32**, and the rest is sent to the secondary-side flow path **35a** of the cascade heat exchanger **35** through the first switching valve **22a** of the secondary-side switching mechanism **22** and the third pipe **25**.

(137) Then, the high-pressure refrigerant sent to the secondary-side first connection pipe **8** is sent to the first branch pipe **63c**. The high-pressure refrigerant sent to the first branch pipe **63c** is sent to the utilization-side heat exchanger **52c** in the utilization unit **3c** through the first regulating valve **66c** and the junction pipe **62c**.

(138) Then, the high-pressure refrigerant sent to the utilization-side heat exchanger **52c** exchanges heat with indoor air supplied by the indoor fan **53c** in the utilization-side heat exchanger **52c**. The refrigerant flowing in the utilization-side heat exchanger **52c** thus radiates heat. The indoor air is heated and is supplied into the indoor space, and the utilization unit **3c** performs the heating operation. The refrigerant having radiated heat in the utilization-side heat exchanger **52c** flows in the second utilization pipe **56c**, and the flow rate of the refrigerant is adjusted at the utilization-side expansion valve **51c**. After that, the refrigerant having flowed through the second connecting tube **16c** is sent to the third branch pipe **61c** in the branch unit **6c**.

(139) Then, the refrigerant sent to the third branch pipe **61c** is sent to the secondary-side third connection pipe **7**.

(140) The high-pressure refrigerant sent to the secondary-side flow path **35a** of the cascade heat exchanger **35** exchanges heat with the primary-side refrigerant flowing in the primary-side flow path **35b** in the cascade heat exchanger **35** to radiate heat. The flow rate of the secondary-side

refrigerant having radiated heat in the cascade heat exchanger **35** is adjusted at the cascade expansion valve **36**, and then the secondary-side refrigerant flows into the secondary-side receiver **45**. A part of the refrigerant having flowed out of the secondary-side receiver **45** branches into the secondary-side subcooling circuit **48**, is decompressed at the secondary-side subcooling expansion valve **48a**, and then joins into the suction flow path **23**. In the secondary-side subcooling heat exchanger **47**, another part of the refrigerant having flowed out of the secondary-side receiver **45** is cooled by the refrigerant flowing in the secondary-side subcooling circuit **48**, is then sent to the secondary-side third connection pipe **7** through the third shutoff valve **31**, and joins the refrigerant having radiated heat in the utilization-side heat exchanger **52c**.

(141) Then, the refrigerant having joined in the secondary-side third connection pipe **7** is branched into two portions to be sent to the third branch pipes **61a** and **61b** of the branch units **6a** and **6b**. Thereafter, the refrigerant having flowed in the second connecting tubes **16a** and **16b** is sent to the second utilization pipes **56a** and **56b** of the first and second utilization units **3a** and **3b**. The refrigerant flowing in the second utilization pipes **56a** and **56b** passes through the utilization-side expansion valves **51a** and **51b** in the utilization units **3a** and **3b**.

(142) The refrigerant having passed through the utilization-side expansion valves **51a** and **51b** whose opening degrees are adjusted exchanges heat with indoor air supplied by the indoor fans **53a** and **53b** in the utilization-side heat exchangers **52a** and **52b**. The refrigerant flowing in the utilization-side heat exchangers **52a** and **52b** is thus evaporated into a low-pressure gas refrigerant. The indoor air is cooled and is supplied into the indoor space. The indoor space is thus cooled. The low-pressure gas refrigerant evaporated in the utilization-side heat exchangers **52a** and **52b** is sent to the junction pipes **62a** and **62b** of the first and second branch units **6a** and **6b**.

(143) The low-pressure gas refrigerant sent to the junction pipes **62a** and **62b** is sent to the secondary-side second connection pipe **9** via the second regulating valves **67a** and **67b** and the second branch pipes **64a** and **64b**, to join.

(144) The low-pressure gas refrigerant sent to the secondary-side second connection pipe **9** is returned to the suction side of the secondary-side compressor **21** via the second shutoff valve **33**, the second pipe **29**, the suction flow path **23**, and the secondary-side accumulator **30**.

(145) Motion during the cooling main operation is performed in such a manner.

(9-4) Heating Main Operation

(146) During the heating main operation, for example, the utilization-side heat exchangers **52a** and **52b** in the utilization units **3a** and **3b** function as refrigerant radiators, and the utilization-side heat exchanger **52c** functions as a refrigerant evaporator. In the heating main operation, the cascade heat exchanger **35** functions as an evaporator for the secondary-side refrigerant. In the heating main operation, the primary-side refrigerant circuit **5a** and the secondary-side refrigerant circuit **10** of the refrigeration cycle apparatus **1** are configured as illustrated in FIG. **6**. Arrows attached to the primary-side refrigerant circuit **5a** and arrows attached to the secondary-side refrigerant circuit **10** in FIG. **6** indicate flows of the refrigerant during the heating main operation.

(147) Specifically, in the primary-side unit **5**, the primary-side switching mechanism **72** is switched to a sixth operating state to cause the cascade heat exchanger **35** to function as a radiator for the primary-side refrigerant. The sixth operating state of the primary-side switching mechanism **72** corresponds to a connection state depicted by broken lines in the primary-side switching mechanism **72** in FIG. **6**. Accordingly, in the primary-side unit **5**, the primary-side refrigerant discharged from the primary-side compressor **71**, having passed through the primary-side switching mechanism **72** and the first gas shutoff valve **109** passes through the primary-side second connection pipe **112** and the second gas shutoff valve **107** and is sent to the primary-side flow path **35b** of the cascade heat exchanger **35**. The refrigerant flowing in the primary-side flow path **35b** of the cascade heat exchanger **35** is condensed by exchanging heat with the secondary-side refrigerant flowing in the secondary-side flow path **35a**. When flowing in the second refrigerant pipe **114**, the primary-side refrigerant condensed in the cascade heat exchanger **35** passes through the primary-

side second expansion valve **102** controlled to the fully opened state. Then, the primary-side refrigerant flows through the second liquid shutoff valve **106**, the primary-side first connection pipe **111**, the first liquid shutoff valve **108**, and the primary-side subcooling heat exchanger **103** in that order, and is decompressed by the primary-side first expansion valve **76**. During the heating main operation, the primary-side subcooling expansion valve **104a** is controlled to the closed state. Accordingly, the refrigerant does not flow into the primary-side subcooling circuit **104** and does not exchange heat in the primary-side subcooling heat exchanger **103**. The valve opening degree of the primary-side first expansion valve **76** is controlled such that, for example, the degree of superheating of the refrigerant sucked into the primary-side compressor **71** satisfies a predetermined condition. The refrigerant decompressed by the primary-side first expansion valve **76** evaporates by exchanging heat with outdoor air supplied from the primary-side fan **75** in the primary-side heat exchanger **74**, passes through the primary-side switching mechanism **72** and the primary-side accumulator **105**, and is sucked into the primary-side compressor **71**.

(148) In the cascade unit **2**, the secondary-side switching mechanism **22** is switched to the second connection state. In the second connection state of the secondary-side switching mechanism **22**, the discharge flow path **24** and the first pipe **28** are connected by the second switching valve **22b**, and the third pipe **25** and the suction flow path **23** are connected by the first switching valve **22a**. The cascade heat exchanger **35** thus functions as an evaporator for the secondary-side refrigerant. The opening degree of the cascade expansion valve **36** is adjusted. In the first to third branch units **6a**, **6b**, and **6c**, the first regulating valves **66a** and **66b** and the second regulating valve **67c** are controlled to the opened state, and the first regulating valve **66c** and the second regulating valves **67a** and **67b** are controlled to the closed state. Accordingly, the utilization-side heat exchangers **52a** and **52b** in the utilization units **3a** and **3b** function as refrigerant radiators, and the utilization-side heat exchanger **52c** in the utilization unit **3c** functions as a refrigerant evaporator. The utilization-side heat exchanger **52c** in the utilization unit **3c** and the suction side of the secondary-side compressor **21** in the cascade unit **2** are connected via the first utilization pipe **57c**, the first connecting tube **15c**, the junction pipe **62c**, the second branch pipe **64c**, and the secondary-side second connection pipe **9**. The utilization-side heat exchangers **52a** and **52b** in the utilization units **3a** and **3b** and the discharge side of the secondary-side compressor **21** in the cascade unit **2** are connected via the discharge flow path **24**, the first pipe **28**, the secondary-side first connection pipe **8**, the first branch pipes **63a** and **63b**, the junction pipes **62a** and **62b**, the first connecting tubes **15a** and **15b**, and the first utilization pipes **57a** and **57b**. The secondary-side subcooling expansion valve **48a** and the bypass expansion valve **46a** are controlled to the closed state. In the utilization units **3a**, **3b**, and **3c**, the opening degrees of the utilization-side expansion valves **51a**, **51b**, and **51c** are adjusted.

(149) In the heating main operation, the secondary-side refrigerant circuit **10** controls capacity, for example, by controlling the frequency of the secondary-side compressor **21** so as to process a load in a heat exchanger functioning as a radiator for the secondary-side refrigerant between the utilization-side heat exchangers **52a**, **52b**, and **52c**. As a result, in the heating main operation, the secondary-side refrigerant discharged from the secondary-side compressor **21** is controlled to be in the critical state exceeding the critical pressure. The primary-side refrigerant circuit **5a** controls capacity, for example, by controlling the frequency of the primary-side compressor **71** such that condensation temperature of the primary-side refrigerant in the primary-side flow path **35b** of the cascade heat exchanger **35** becomes a predetermined primary-side condensation target temperature.

(150) In such a secondary-side refrigerant circuit **10**, the secondary-side high-pressure refrigerant compressed and discharged by the secondary-side compressor **21** is sent to the secondary-side first connection pipe **8** through the second switching valve **22b** of the secondary-side switching mechanism **22**, the first pipe **28**, and the first shutoff valve **32**.

(151) The high-pressure refrigerant sent to the secondary-side first connection pipe **8** is branched into two portions to be sent to the first branch pipes **63a** and **63b** of the first branch unit **6a** and the

second branch unit **6b** connected to the first utilization unit **3a** and the second utilization unit **3b** which are utilization units in operation. The high-pressure refrigerant sent to the first branch pipes **63a** and **63b** is sent to the utilization-side heat exchangers **52a** and **52b** in the first utilization unit **3a** and the second utilization unit **3b** via the first regulating valves **66a** and **66b**, the junction pipes **62a** and **62b**, and the first connecting tubes **15a** and **15b**.

(152) The high-pressure refrigerant sent to the utilization-side heat exchangers **52a** and **52b** exchanges heat with indoor air supplied by the indoor fans **53a** and **53b** in the utilization-side heat exchangers **52a** and **52b**. The refrigerant flowing in the utilization-side heat exchangers **52a** and **52b** thus radiates heat. The indoor air is heated and supplied into the indoor space. The indoor space is thus heated. The refrigerant having radiated heat in the utilization-side heat exchangers **52a** and **52b** flows in the second utilization pipes **56a** and **56b**, and passes through the utilization-side expansion valves **51a** and **51b** whose opening degree is adjusted. The secondary-side refrigerant that has passed through the utilization-side expansion valves **51a** and **51b** has the critical pressure or less. Thereafter, the refrigerant having flowed through the second connecting tubes **16a** and **16b** is sent to the secondary-side third connection pipe **7** via the third branch pipes **61a** and **61b** of the branch units **6a** and **6b**.

(153) A part of the refrigerant sent to the secondary-side third connection pipe **7** is sent to the third branch pipe **61c** of the branch unit **6c**, and the rest flows toward the third shutoff valve **31**.

(154) Then, the refrigerant sent to the third branch pipe **61c** flows in the second utilization pipe **56c** of the utilization unit **3c** via the second connecting tube **16c**, and is sent to the utilization-side expansion valve **51c**.

(155) The refrigerant having passed through the utilization-side expansion valve **51c** whose opening degree is adjusted exchanges heat with indoor air supplied by the indoor fan **53c** in the utilization-side heat exchanger **52c**. The refrigerant flowing in the utilization-side heat exchanger **52c** is thus evaporated into a low-pressure gas refrigerant. The indoor air is cooled and is supplied into the indoor space. The indoor space is thus cooled. The low-pressure gas refrigerant evaporated in the utilization-side heat exchanger **52c** passes through the first utilization pipe **57c** and the first connecting tube **15c** to be sent to the junction pipe **62c**.

(156) The low-pressure gas refrigerant sent to the junction pipe **62c** is sent to the secondary-side second connection pipe **9** through the second regulating valve **67c** and the second branch pipe **64c**.

(157) The low-pressure gas refrigerant sent to the secondary-side second connection pipe **9** is returned to the suction side of the secondary-side compressor **21** via the second shutoff valve **33**, the second pipe **29**, the suction flow path **23**, and the secondary-side accumulator **30**.

(158) The refrigerant flowing toward the third shutoff valve **31** is sent to the cascade expansion valve **36**. The refrigerant sent to the cascade expansion valve **36** passes through the cascade expansion valve **36** whose opening degree is adjusted, and then exchanges heat with the primary-side refrigerant flowing in the primary-side flow path **35b** in the secondary-side flow path **35a** of the cascade heat exchanger **35**. As a result, the refrigerant flowing in the secondary-side flow path **35a** of the cascade heat exchanger **35** evaporates to become a low-pressure gas refrigerant, and is sent to the first switching valve **22a** of the secondary-side switching mechanism **22**. The low-pressure gas refrigerant sent to the first switching valve **22a** of the secondary-side switching mechanism **22** joins the low-pressure gas refrigerant evaporated in the utilization-side heat exchanger **52c** in the suction flow path **23**. The refrigerant thus joined is returned to the suction side of the secondary-side compressor **21** via the secondary-side accumulator **30**.

(159) Motion during the heating main operation is performed in such a manner.

(9-5) Oil Return Operation

(160) When a predetermined oil return condition is satisfied during the heating operation and the heating main operation, the operation is performed with the connection state of the secondary-side refrigerant circuit **10** temporarily set as the connection state during the cooling operation to perform the oil return operation. As a result, the refrigerating machine oil retained at a lower end of the

vessel body **90** of the secondary-side receiver **45** can be extracted from the secondary-side receiver **45** through the second refrigerant pipe **97** of the fifth pipe **27** extending from a lower end of the secondary-side receiver **45**.

(161) The predetermined oil return condition is not limited, and for example, the heating operation and the heating main operation may be performed for a predetermined time. An end of the oil return operation is not limited, and for example, the oil return operation may be performed for a predetermined time.

(10) Secondary-Side Receiver

(162) FIG. **7** is a schematic configuration diagram of the secondary-side receiver **45**.

(163) The secondary-side receiver **45** includes the vessel body **90**, a first refrigerant pipe **96**, a second refrigerant pipe **97**, and a third refrigerant pipe **98**.

(164) The vessel body **90** is a substantially cylindrical vessel having an internal volume corresponding to the amount of refrigerant filled in the secondary-side refrigerant circuit **10**, and temporarily reserves the refrigerant flowing in the secondary-side refrigerant circuit **10**. The vessel body **90** includes an upper part **91**, a lower peripheral surface part **92**, and a bottom part **93**, which are welded and fixed to each other. The upper part **91** has a cylindrical shape in which an internal space is formed so as to open downward. The lower peripheral surface part **92** is disposed under the upper part **91** and has a cylindrical shape penetrating in an up-down direction. The bottom part **93** is a columnar member whose thickness direction is the up-down direction. The bottom part **93** has an upper surface **93a** facing upward, toward a space inside the secondary-side receiver **45**, and a lower surface **93b** facing downward, toward a space outside the secondary-side receiver **45**. In the present embodiment, both the upper surface **93a** and the lower surface **93b** extend in a horizontal plane. The bottom part **93** has a portion penetrating in the up-down direction at a center in plan view, and an end of the second refrigerant pipe **97** on a side connected to the secondary-side receiver **45** is inserted from below. The bottom part **93** and the second refrigerant pipe **97** are welded to each other.

(165) In the present embodiment, the first refrigerant pipe **96** is, for example, a pipe extending laterally from a part of a peripheral surface of the vessel body **90**, and constitutes a part of the fourth pipe **26** in the secondary-side refrigerant circuit **10**. The first refrigerant pipe **96** is welded to the upper part **91** of the vessel body **90**. The first refrigerant pipe **96** has a portion extending downward inside the secondary-side receiver **45**, and an end located inside the secondary-side receiver **45** has an opening **96a** formed to be inclined with respect to an axial direction in which the pipe extends. The end of the first refrigerant pipe **96** in the secondary-side receiver **45** is in contact with the upper surface **93a** of the bottom part **93** or extends to a position in front of the upper surface **93a** of the bottom part **93**.

(166) The second refrigerant pipe **97** is a pipe extending downward from the bottom part **93** of the vessel body **90**, and constitutes a part of the fifth pipe **27** in the secondary-side refrigerant circuit **10**. The second refrigerant pipe **97** is welded to the bottom part **93** in a state of being inserted into a penetrating portion in the up-down direction formed in the bottom part **93**. In the present embodiment, the upper end **97a** of the second refrigerant pipe **97** is disposed at a height position equal to or lower than a height position of a portion of the upper surface **93a** of the bottom part **93** where the second refrigerant pipe **97** is connected. Specifically, in the present embodiment, the upper end **97a** of the second refrigerant pipe **97** is disposed on the same horizontal plane as the upper surface **93a** of the bottom part **93**.

(167) The third refrigerant pipe **98** is a pipe extending laterally from a part of the peripheral surface of the vessel body **90**, and constitutes a part of the bypass circuit **46** in the secondary-side refrigerant circuit **10**. The third refrigerant pipe **98** is welded to the upper part **91** of the vessel body **90**.

(11) Characteristics of Embodiment

(168) In the secondary-side refrigerant circuit **10** of the refrigeration cycle apparatus **1** according to

the present embodiment, the carbon dioxide refrigerant and the PAG oil which are incompatible with each other are used. Here, the PAG oil has a higher density than the carbon dioxide refrigerant in a use condition in the secondary-side refrigerant circuit **10**. Therefore, in the secondary-side receiver **45**, the PAG oil is likely to be located below the carbon dioxide refrigerant.

(169) In the secondary-side receiver **45** according to the present embodiment, the second refrigerant pipe **97** is provided so as to penetrate the bottom part **93** in the up-down direction. The upper end **97a** of the second refrigerant pipe **97** is disposed at the same height position as the upper surface **93a** of the bottom part **93** or at a height position lower than the upper surface **93a** of the bottom part **93**. As a result, the PAG oil flowing into the secondary-side receiver **45** via the first refrigerant pipe **96** and located below carbon dioxide refrigerant passes through the second refrigerant pipe **97** and is efficiently sent out of the secondary-side receiver **45**. Therefore, the refrigerating machine oil is prevented from being continuously reserved in the secondary-side receiver **45**.

(170) In the refrigeration cycle apparatus **1** according to the present embodiment, the carbon dioxide refrigerant discharged from the secondary-side compressor **21** of the secondary-side refrigerant circuit **10** is not in the supercritical state by exchanging heat with the primary-side refrigerant flowing in the primary-side refrigerant circuit **5a** in the cascade heat exchanger **35** or by controlling the valve opening degree of the cascade expansion valve **36**, and is sent to the secondary-side receiver **45**. Therefore, even if the carbon dioxide refrigerant and the refrigerating machine oil are phase-separated in the secondary-side receiver **45**, it is possible to prevent the refrigerating machine oil from continuously accumulating in the secondary-side receiver **45** while avoiding the supercritical state in which the behavior tends to be unstable.

(12) Other Embodiments

(12-1) Another Embodiment A

(171) In the above embodiment, as an example, a case has been described where the upper surface **93a** of the bottom part **93** of the secondary-side receiver **45** is a horizontal plane.

(172) Alternatively, for example, as illustrated in FIG. **8**, the bottom part **93** of the secondary-side receiver **45** may have an upper curved surface **193a** that is curved so as to gently protrude downward and constitutes an inner bottom surface of the secondary-side receiver **45**, instead of the upper surface **93a** according to the above embodiment. The upper curved surface **193a** has a curved shape that gradually descends toward the upper end **97a** of the second refrigerant pipe **97**. The upper end **97a** of the second refrigerant pipe **97** is located at a lowermost end of the upper curved surface **193a**.

(173) In this case, since the refrigerating machine oil located on the upper curved surface **193a** is guided to the upper end **97a** of the second refrigerant pipe **97** by a weight of the refrigerating machine oil, the refrigerating machine oil can be efficiently discharged from the secondary-side receiver **45**.

(12-2) Another Embodiment B

(174) In the another embodiment A, as an example, a case has been described where the bottom part **93** of the secondary-side receiver **45** has the upper curved surface **193a**.

(175) Alternatively, for example, as illustrated in FIG. **9**, the bottom part **93** of the secondary-side receiver **45** may have a lower curved surface **193b** that is curved so as to protrude downward similarly to the upper curved surface **193a** and constitutes an outer bottom surface of the secondary-side receiver **45**, instead of the lower surface **93b**.

(176) In this case, not only the effects of the above embodiment and the another embodiment A can be obtained, but also the thickness of the bottom part **93** in the up-down direction can be made constant, and a pressure resistance strength of the secondary-side receiver **45** can be easily secured.

(12-3) Another Embodiment C

(177) In the above embodiment, as an example, a case has been described where the height position of the upper end **97a** of the second refrigerant pipe **97** is equal to or lower than the height position

of the portion of the upper surface **93a** of the bottom part **93** where the second refrigerant pipe **97** is connected.

(178) However, the height position of the upper end **97a** of the second refrigerant pipe **97** is not limited to the height position as described above.

(179) For example, as illustrated in FIG. **10**, the upper end **197a** of the second refrigerant pipe **197** may be located slightly above a portion of the upper surface **93a** of the bottom part **93** where the second refrigerant pipe **197** is connected. For example, the height position of the upper end **197a** of the second refrigerant pipe **197** may be higher than the portion of the upper surface **93a** of the bottom part **93** where the second refrigerant pipe **197** is connected, and may be lower than a height position 15 mm higher than the portion. The height position of the upper end **197a** of the second refrigerant pipe **197** may be equal to or lower than a height position 10 mm higher than the portion of the upper surface **93a** of the bottom part **93** where the second refrigerant pipe **197** is connected. A difference in a height direction between the upper end **197a** of the second refrigerant pipe **197** and the portion of the upper surface **93a** of the bottom part **93** where the second refrigerant pipe **197** is connected may be 1/100 or less of a height of the vessel body **90** of the secondary-side receiver **45**. Since the upper end **197a** of the second refrigerant pipe **197** protrudes further upward than the upper surface **93a** of the bottom part **93** as described above, a pipe protruding margin can be secured to facilitate manufacturing.

(180) A portion above the upper surface **93a** of the bottom part **93** near the upper end **197a** of the second refrigerant pipe **197** may be flared such that a flow path area increases upward.

(12-4) Another Embodiment D

(181) In the above embodiment, as an example, a case has been described where the first refrigerant pipe **96** is provided so as to extend laterally from a part of a peripheral surface of the secondary-side receiver **45**.

(182) However, a connection form of the first refrigerant pipe **96** with the secondary-side receiver **45** is not limited to the above case.

(183) For example, as illustrated in FIG. **11**, the first refrigerant pipe **196** may be provided so as to extend downward from the bottom part **93** of the secondary-side receiver **45**. In this case, the upper end **196a** of the first refrigerant pipe **196** may be configured similarly to the upper end **97a** of the second refrigerant pipe **97** according to the above embodiment, or similarly to the upper end **197a** of the second refrigerant pipe **197** of the another embodiment C.

(12-5) Another Embodiment E

(184) In the above embodiment, as an example, a case has been described where the second refrigerant pipe **97** is provided so as to penetrate the bottom part **93** in the up-down direction in the secondary-side receiver **45** as a high-pressure receiver provided at a portion in which the high-pressure refrigerant flows in the secondary-side refrigerant circuit **10**.

(185) Alternatively, for example, as an intermediate-pressure receiver **18** in a refrigerant circuit **210** of an air conditioner **201** illustrated in FIG. **12**, a second refrigerant pipe **19** may be provided so as to penetrate the bottom part in the up-down direction.

(186) The air conditioner **201** is a device that performs a vapor compression refrigeration cycle to condition air in a target space. The air conditioner **201** mainly includes an outdoor unit **202**, an indoor unit **203**, a liquid-side connection pipe **207** and a gas-side connection pipe **208** that connect the outdoor unit **202** and the indoor unit **203**, and a controller **4** that controls motion of the air conditioner **201**. The air conditioner **201** performs a refrigeration cycle in which a refrigerant sealed in the refrigerant circuit **210** is compressed, radiates heat, is decompressed, evaporated, and then compressed again. The air conditioner **201** is filled with a refrigerant and a refrigerating machine oil that is incompatible with the refrigerant and has a higher density than the refrigerant. Specifically, the refrigerant is a carbon dioxide refrigerant, and the refrigerating machine oil is PAG oil.

(187) The outdoor unit **202** is connected to the indoor unit **203** via the liquid-side connection pipe

207 and the gas-side connection pipe **208**, and constitutes a part of the refrigerant circuit **210**. The outdoor unit **202** mainly includes a compressor **221**, a four-way switching valve **222**, an outdoor heat exchanger **235**, a first refrigerant pipe **17**, a first outdoor expansion valve **17a**, the intermediate-pressure receiver **18**, the second refrigerant pipe **19**, a second outdoor expansion valve **19a**, an outdoor fan **11**, a liquid-side shutoff valve **231**, and a gas-side shutoff valve **232**.

(188) The compressor **221** is a device that compresses a low-pressure refrigerant in the refrigeration cycle to a high pressure. The compressor **221** used herein is a closed compressor in which a rotary type, scroll type, or other positive-displacement compression element is driven to rotate by a compressor motor. The compressor motor is for changing a capacity, and an operating frequency can be controlled by an inverter.

(189) By switching the connection state, the four-way switching valve **222** can switch between a connection state in which a cooling operation is performed by connecting a suction side of the compressor **221** and the gas-side shutoff valve **232** while connecting a discharge side of the compressor **221** and the outdoor heat exchanger **235**, and a connection state in which a heating operation is performed by connecting the suction side of the compressor **221** and the outdoor heat exchanger **235** while connecting the discharge side of the compressor **221** and the gas-side shutoff valve **232**.

(190) The outdoor heat exchanger **235** functions as a radiator for the high-pressure refrigerant in the refrigeration cycle during the cooling operation, and functions as an evaporator for the low-pressure refrigerant in the refrigeration cycle during the heating operation. The outdoor heat exchanger **235** includes a plurality of heat transfer fins and a plurality of heat transfer tubes penetrated through and fixed to the plurality of heat transfer fins. The outdoor heat exchanger **235** is an air heat exchanger that causes heat exchange between the refrigerant and air flowing outside.

(191) The outdoor fan **11** sucks outdoor air into the outdoor unit **202**, causes the outdoor air to exchange heat with the refrigerant in the outdoor heat exchanger **235**, and then generates an air flow to be discharged to the outside. The outdoor fan **11** is driven to rotate by an outdoor fan motor.

(192) The first refrigerant pipe **17** connects an end of the outdoor heat exchanger **235** on an opposite side of the compressor **221** and the intermediate-pressure receiver **18**. The first refrigerant pipe **17** is provided with a first outdoor expansion valve **17a** whose valve opening degree is controllable. During the cooling operation, the opening degree of the first outdoor expansion valve **17a** is controlled in order to send the intermediate-pressure refrigerant obtained by decompressing the high-pressure refrigerant to the intermediate-pressure receiver **18**. During the heating operation, the opening degree of the first outdoor expansion valve **17a** is controlled in order to send the low-pressure refrigerant obtained by decompressing the intermediate-pressure refrigerant to the outdoor heat exchanger **235**.

(193) The intermediate-pressure receiver **18** is provided between the first outdoor expansion valve **17a** and the second outdoor expansion valve **19a** in the refrigerant circuit **210**. Note that the details of the specific structure of the intermediate-pressure receiver **18** are similar to those of the secondary-side receiver **45** according to the above embodiment.

(194) The second refrigerant pipe **19** connects the bottom part of the intermediate-pressure receiver **18** and the liquid-side shutoff valve **231**. The second refrigerant pipe **19** is provided with a second outdoor expansion valve **19a** whose valve opening degree is controllable. During the heating operation, the opening degree of the second outdoor expansion valve **19a** is controlled in order to send the intermediate-pressure refrigerant obtained by decompressing the high-pressure refrigerant to the intermediate-pressure receiver **18**. During the cooling operation, the opening degree of the second outdoor expansion valve **19a** is controlled in order to send the low-pressure refrigerant obtained by decompressing the intermediate-pressure refrigerant to the indoor heat exchanger **252**.

(195) The liquid-side shutoff valve **231** is a manual valve disposed at a connecting part of the outdoor unit **202** with the liquid-side connection pipe **207**. The gas-side shutoff valve **232** is a manual valve disposed at a connecting part of the outdoor unit **202** with the gas-side connection

pipe **208**.

(196) The outdoor unit **202** includes an outdoor unit control unit **4a** that controls motion of each element constituting the outdoor unit **202**. The indoor unit **203** is installed on, for example, an indoor wall surface as a target space. The indoor unit **203** is connected to the outdoor unit **202** via the liquid-side connection pipe **207** and the gas-side connection pipe **208**, and constitutes a part of the refrigerant circuit **210**. The indoor unit **203** includes the indoor heat exchanger **252**, an indoor fan **253**, and the like.

(197) The indoor heat exchanger **252** has a liquid side connected to the liquid-side connection pipe **207** and a gas side end connected to the gas-side connection pipe **208**. The indoor heat exchanger **252** functions as an evaporator for the low-pressure refrigerant in the refrigeration cycle during the cooling operation, and functions as a condenser for the high-pressure refrigerant in the refrigeration cycle during the heating operation. The indoor heat exchanger **252** includes a plurality of heat transfer fins and a plurality of heat transfer tubes penetrated through and fixed to the plurality of heat transfer fins.

(198) The indoor fan **253** sucks indoor air into the indoor unit **203**, causes the indoor air to exchange heat with the refrigerant in the indoor heat exchanger **252**, and then generates an air flow to be discharged to the outside.

(199) The indoor unit **203** includes an indoor unit control unit **4b** that controls motion of each element constituting the indoor unit **203**.

(200) As in the above embodiment, the intermediate-pressure receiver **18** described above can efficiently discharge the refrigerating machine oil which is incompatible with the carbon dioxide refrigerant, has a higher density than the carbon dioxide refrigerant, and tends to be located below the carbon dioxide refrigerant through the second refrigerant pipe **19**.

(12-6) Another Embodiment F

(201) In the above embodiment, description has been made by exemplifying the refrigeration cycle apparatus **1** in which one cascade unit **2** is connected to one primary-side unit **5**.

(202) Alternatively, as illustrated in FIG. **13**, for example, by connecting are a plurality of cascade units, namely, a first cascade unit **2a**, a second cascade unit **2b**, and a third cascade unit **2c**, in parallel to each other to one primary-side unit **5**, the refrigeration cycle apparatus **1** may include a first secondary-side refrigerant circuit **10a** including a first cascade circuit **12a**, a second secondary-side refrigerant circuit **10b** including a second cascade circuit **12b**, and a third secondary-side refrigerant circuit **10c** including a third cascade circuit **12c**. Note that, in FIG. **13**, an internal structure of each of the first cascade unit **2a**, the second cascade unit **2b**, and the third cascade unit **2c** is similar to that of the cascade unit **2** according to the above embodiment, and thus only a part of each cascade unit is illustrated.

(203) Although not illustrated, each of the first cascade unit **2a**, the second cascade unit **2b**, and the third cascade unit **2c** is connected to the plurality of branch units **6a**, **6b**, and **6c** and the plurality of utilization units **3a**, **3b**, and **3c** as in the above embodiment. Specifically, the first cascade unit **2a** is connected to a plurality of branch units and utilization units via a secondary-side third connection pipe **7a**, a secondary-side first connection pipe **8a**, and a secondary-side second connection pipe **9a**. The second cascade unit **2b** is connected, via a secondary-side third connection pipe **7b**, a secondary-side first connection pipe **8b**, and a secondary-side second connection pipe **9b**, to a plurality of branch units and utilization units different from those connected to the first cascade unit **2a**. The third cascade unit **2c** is connected, via a secondary-side third connection pipe **7c**, a secondary-side first connection pipe **8c**, and a secondary-side second connection pipe **9c**, to another plurality of branch units and utilization units different from those connected to the first cascade unit **2a** and different from those connected to the second cascade unit **2b**.

(204) Here, the primary-side unit **5** and the first cascade unit **2a** are connected via a primary-side first connection pipe **111a** and a primary-side second connection pipe **112a**. The primary-side unit **5** and the second cascade unit **2b** are connected via a primary-side first connection pipe **111b**

branched from the primary-side first connection pipe **111a** and a primary-side second connection pipe **112b** branched from the primary-side second connection pipe **112a**. The primary-side unit **5** and the third cascade unit **2c** are connected via a primary-side first connection pipe **111c** branched from the primary-side first connection pipe **111a** and a primary-side second connection pipe **112c** branched from the primary-side second connection pipe **112a**.

(205) Here, each of the first cascade unit **2a**, the second cascade unit **2b**, and the third cascade unit **2c** includes a primary-side second expansion valve **102** whose opening degree is controlled by the first cascade unit **2a**, the second cascade unit **2b**, and the third cascade unit **2c**. Furthermore, a first cascade-side control unit **20a** included in the first cascade unit **2a**, a second cascade-side control unit **20b** included in the second cascade unit **2b**, and a third cascade-side control unit **20c** included in the third cascade unit **2c** control the opening degree of the corresponding primary-side second expansion valve **102**. Similarly to the above embodiment, each of the first cascade-side control unit **20a**, the second cascade-side control unit **20b**, and the third cascade-side control unit **20c** controls the valve opening degree of the corresponding primary-side second expansion valve **102** on the basis of conditions of the first cascade circuit **12a**, the second cascade circuit **12b**, and the third cascade circuit **12c** controlled by the first cascade-side control unit **20a**, the second cascade-side control unit **20b**, and the third cascade-side control unit **20c**. As a result, the primary-side refrigerant flowing through the primary-side refrigerant circuit **5a** is controlled to have a flow rate of the primary-side refrigerant in the primary-side first connection pipe **111a** and the primary-side second connection pipe **112a**, a flow rate of the primary-side refrigerant in the primary-side first connection pipe **111b** and the primary-side second connection pipe **112b**, and a flow rate of the primary-side refrigerant in the primary-side first connection pipe **111c** and the primary-side second connection pipe **112c** so as to correspond to a difference in loads in the first secondary-side refrigerant circuit **10a**, the second secondary-side refrigerant circuit **10b**, and the third secondary-side refrigerant circuit **10c**.

(12-7) Another Embodiment G

(206) In the above embodiment, R32 or R410A is exemplified as the refrigerant used in the primary-side refrigerant circuit **5a**, and carbon dioxide is exemplified as the refrigerant used in the secondary-side refrigerant circuit **10**.

(207) Alternatively, the refrigerant provided in the primary-side refrigerant circuit **5a** should not be limited, and examples of the refrigerant include HFC-32, an HFO refrigerant, a refrigerant obtained by mixing HFC-32 and the HFO refrigerant, carbon dioxide, ammonia, and propane.

(208) Furthermore, instead of the primary-side refrigerant circuit **5a** in which the refrigerant flows, a heat medium circuit in which a heat medium such as water or brine flows may be used. In this case, the heat medium circuit may include a heat source that functions as a heat source or a cold source, and a pump for circulating the heat medium. In this case, the flow rate can be adjusted by the pump, and the amount of heat can be controlled by the heat source or the cold source.

(209) The refrigerant provided in the secondary-side refrigerant circuit **10** should not be limited, and examples of the refrigerant include HFC-32, an HFO refrigerant, a refrigerant obtained by mixing HFC-32 and the HFO refrigerant, carbon dioxide, ammonia, and propane.

(210) Note that examples of the HFO refrigerant include HFO-1234yf and HFO-1234ze.

(211) The same refrigerant or different refrigerants may be used in the primary-side refrigerant circuit **5a** and the secondary-side refrigerant circuit **10**. Preferably, the refrigerant used in the secondary-side refrigerant circuit **10** has at least one of lower global warming potential (GWP), lower ozone depletion potential (ODP), lower flammability, or lower toxicity than the refrigerant used in the primary-side refrigerant circuit **5a**. Here, the flammability can be compared in accordance with classifications related to ASHRAE 34 flammability, for example. Note that the toxicity can be compared, for example, in accordance with classifications related to ASHRAE 34 safety grade. In particular, when an overall content volume of the secondary-side refrigerant circuit **10** is larger than an overall content volume of the primary-side refrigerant circuit **5a**, by using the

refrigerant lower than the refrigerant in the primary-side refrigerant circuit **5a** in at least one of the global warming potential (GWP), the ozone depletion potential (ODP), the flammability, or the toxicity in the secondary-side refrigerant circuit **10**, adverse effects when a leak occurs can be reduced.

(12-8) Others

(212) In some embodiments, the first refrigerant pipe may introduce the refrigerant into the vessel body, and the second refrigerant pipe may derive the refrigerant from the vessel body.

(213) That the refrigerating machine oil is incompatible with the refrigerant means that the refrigerant and the refrigerating machine oil are separated from each other without becoming a uniform layer of liquid in the refrigerant vessel under a use environment of the refrigerant and the refrigerating machine oil.

(214) That the density of the refrigerating machine oil is higher than the density of the refrigerant means that the density of the refrigerating machine oil is higher than the density of the refrigerant in a liquid state under the use environment of the refrigerant and the refrigerating machine oil.

(215) The height position of the upper end of the second refrigerant pipe is preferably equal to or lower than a position 10 mm higher than the lowermost end of the inner peripheral surface of the bottom part of the vessel body.

SUPPLEMENTARY NOTE

(216) Although the embodiments of the present disclosure have been described above, it will be understood that various changes in form and details can be made without departing from the gist and scope of the present disclosure described in the claims.

REFERENCE SIGNS LIST

(217) **1**: refrigeration cycle apparatus **2**: cascade unit **2x**: cascade casing **3a**: first utilization unit **3b**: second utilization unit **3c**: third utilization unit **5**: primary-side unit **5a**: primary-side refrigerant circuit (first refrigerant circuit) **10**: secondary-side refrigerant circuit (refrigerant circuit, second refrigerant circuit) **12**: cascade circuit **13a**, **13b**, **13c**: utilization circuit **18**: intermediate-pressure receiver (refrigerant vessel) **20**: cascade-side control unit **21**: secondary-side compressor (compressor) **21a**: compressor motor **22**: secondary-side switching mechanism **22a**: first switching valve **22b**: second switching valve **22x**: discharge-side connection portion **22y**: suction-side connection portion **23**: suction flow path (refrigerant flow path) **24**: discharge flow path **25**: third pipe **26**: fourth pipe **27**: fifth pipe **28**: first pipe **29**: second pipe **30**: secondary-side accumulator **34**: oil separator **35**: cascade heat exchanger **35a**: secondary-side flow path **35b**: primary-side flow path **36**: cascade expansion valve **45**: secondary-side receiver (refrigerant vessel) **46**: bypass circuit (refrigerant flow path) **46a**: bypass expansion valve **47**: secondary-side subcooling heat exchanger **48**: secondary-side subcooling circuit **48a**: secondary-side subcooling expansion valve **50a-c**: utilization-side control unit **51a-c**: utilization-side expansion valve **52a-c**: utilization-side heat exchanger **53a-c**: indoor fan **60a**, **60b**, **60c**: branch unit control unit **66a**, **66b**, **66c**: first regulating valve **67a**, **67b**, **67c**: second regulating valve **68a**, **68b**, **68c**: check valve **69a**, **69b**, **69c**: bypass pipe **70**: primary-side control unit **71**: primary-side compressor **72**: primary-side switching mechanism **74**: primary-side heat exchanger **76**: primary-side first expansion valve **80**: control unit **90**: vessel body **91**: upper part **92**: lower peripheral surface portion **93**: bottom part **93a**: upper surface **93b**: lower surface **96**: first refrigerant pipe **97**: second refrigerant pipe **97a**: upper end **98**: third refrigerant pipe **102**: primary-side second expansion valve **103**: primary-side subcooling heat exchanger **104**: primary-side subcooling circuit **104a**: primary-side subcooling expansion valve **105**: primary-side accumulator **111**: primary-side first connection pipe **112**: primary-side second connection pipe **113**: first refrigerant pipe **114**: second refrigerant pipe **193a**: upper curved surface (curved portion) **193b**: lower curved surface **196**: first refrigerant pipe **197**: second refrigerant pipe **197a**: upper end **210**: refrigerant circuit

CITATION LIST

Claims

1. A refrigerant vessel that reserves a refrigerant containing a carbon dioxide refrigerant in a refrigerant circuit in which the refrigerant and a refrigerating machine oil circulate, the refrigerant vessel comprising: a vessel body including a cylindrical member and a bottom member that is welded and fixed to the cylindrical member and has a vertical direction as the thickness direction; a first refrigerant pipe connected to the vessel body; a second refrigerant pipe connected to the vessel body; and a third refrigerant pipe connected to the vessel body, wherein the bottom member has a concave curved surface facing a space inside the refrigerant vessel and is concaved downwardly, the refrigerating machine oil is incompatible with the refrigerant and has a density higher than a density of the refrigerant, an opening of the first refrigerant pipe is above an opening of the second refrigerant pipe and below an opening of the third refrigerant pipe in the vessel body, the first refrigerant pipe includes a portion that extends in a downward direction within the vessel body and has an opening formed to be inclined with respect to an axial direction in which the first refrigerant pipe extends, the second refrigerant pipe extends in the downward direction from the bottom member of the vessel body, an opening of the second refrigerant pipe is offset from the opening of the first refrigerant pipe in a second direction perpendicular to the downward direction such that when the refrigerating machine oil is inserted into the vessel body via the opening of the first refrigerant pipe, the refrigerating machine oil guided by the concave curved surface of the bottom member to the opening of the second refrigerant pipe, the opening of the first refrigerant pipe being inclined away from the opening of the second refrigerant pipe, and the bottom member includes an indented cylindrical member with an indented cylindrical shape, with the vertical direction as the thickness direction, and includes the downward concave curved surface facing a space inside the refrigerant vessel forming an indent in the indented cylindrical shape.
2. The refrigerant vessel according to claim 1, wherein a height position of an upper end of the second refrigerant pipe is equal to or lower than a lowermost end of an inner peripheral surface of the bottom member of the vessel body.
3. The refrigerant vessel according to claim 2, wherein the refrigerant vessel is a high-pressure receiver in which a high-pressure refrigerant is reserved, the high-pressure receiver being provided at a portion of the refrigerant circuit in which the high-pressure refrigerant flows.
4. The refrigerant vessel according to claim 2, wherein the refrigerant circuit includes a portion in which a high-pressure refrigerant, a low-pressure refrigerant, and an intermediate-pressure refrigerant flow, and the refrigerant vessel is an intermediate-pressure receiver in which the intermediate-pressure refrigerant is reserved, the intermediate-pressure receiver being provided at a portion of the refrigerant circuit in which the intermediate-pressure refrigerant flows.
5. The refrigerant vessel according to claim 2, wherein the refrigerant circuit includes a compressor, and the refrigerant vessel further includes a third refrigerant pipe that is connected to the vessel body and guides the refrigerant in the vessel body to a suction side of the compressor.
6. The refrigerant vessel according to claim 2, wherein the refrigerating machine oil includes polyalkylene glycol oil.
7. A refrigeration cycle apparatus comprising the refrigerant circuit including the refrigerant vessel according to claim 2.
8. The refrigerant vessel according to claim 1, wherein the refrigerant vessel is a high-pressure receiver in which a high-pressure refrigerant is reserved, the high-pressure receiver being provided at a portion of the refrigerant circuit in which the high-pressure refrigerant flows.
9. The refrigerant vessel according to claim 8, wherein the refrigerant circuit includes a compressor, and the refrigerant vessel further includes a third refrigerant pipe that is connected to

the vessel body and guides the refrigerant in the vessel body to a suction side of the compressor.

10. The refrigerant vessel according to claim 8, wherein the refrigerating machine oil includes polyalkylene glycol oil.

11. A refrigeration cycle apparatus comprising the refrigerant circuit including the refrigerant vessel according to claim 8.

12. The refrigerant vessel according to claim 1, wherein the refrigerant circuit includes a portion in which a high-pressure refrigerant, a low-pressure refrigerant, and an intermediate-pressure refrigerant flow, and the refrigerant vessel is an intermediate-pressure receiver in which the intermediate-pressure refrigerant is reserved, the intermediate-pressure receiver being provided at a portion of the refrigerant circuit in which the intermediate-pressure refrigerant flows.

13. The refrigerant vessel according to claim 12, wherein the refrigerant circuit includes a compressor, and the refrigerant vessel further includes a third refrigerant pipe that is connected to the vessel body and guides the refrigerant in the vessel body to a suction side of the compressor.

14. The refrigerant vessel according to claim 12, wherein the refrigerating machine oil includes polyalkylene glycol oil.

15. The refrigerant vessel according to claim 1, wherein the refrigerant circuit includes a compressor, and the third refrigerant pipe is connected to the vessel body and guides the refrigerant in the vessel body to a suction side of the compressor.

16. The refrigerant vessel according to claim 15, wherein the refrigerating machine oil includes polyalkylene glycol oil.

17. The refrigerant vessel according to claim 1, wherein the refrigerating machine oil includes polyalkylene glycol oil.

18. A refrigeration cycle apparatus comprising the refrigerant circuit including the refrigerant vessel according to claim 1.

19. The refrigeration cycle apparatus according to claim 18, comprising: a first refrigerant circuit in which a refrigerant flows; and a cascade heat exchanger, wherein the refrigerant circuit is provided as a second refrigerant circuit independent of the first refrigerant circuit, the cascade heat exchanger exchanges heat between the refrigerant flowing in the first refrigerant circuit and the refrigerant flowing in the second refrigerant circuit, and the refrigerant in the second refrigerant circuit that has become a liquid refrigerant by being cooled by the refrigerant in the first refrigerant circuit in the cascade heat exchanger is reserved in the refrigerant vessel.

20. A refrigeration cycle apparatus comprising: a first refrigerant circuit in which a first refrigerant flows; and a second refrigerant circuit in which a refrigerating machine oil and a second refrigerant containing a carbon dioxide refrigerant circulate, the second refrigerant circuit is independent of the first refrigerant circuit; and a cascade heat exchanger exchanges heat between the first refrigerant flowing in the first refrigerant circuit and the second refrigerant flowing in the second refrigerant circuit, wherein the first refrigerant circuit has a compressor sucking the first refrigerant passed through the cascade heat exchanger, the second refrigerant circuit has a refrigerant vessel that reserves the second refrigerant and an expansion valve provided between the cascade heat exchanger and the refrigerant vessel, the refrigerant vessel has a vessel body, a first refrigerant pipe connected to the vessel body, and a second refrigerant pipe connected to the vessel body, the refrigerant machine oil is incompatible with the second refrigerant and has a density higher than a density of the second refrigerant, the second refrigerant pipe extends downward from a bottom of the vessel body, the second refrigerant that has become a liquid refrigerant by being cooled by the first refrigerant in the cascade heat exchanger is reserved in the refrigerant vessel, and the compressor or both the compressor and the expansion valve are controlled so that the second refrigerant flowing from the cascade heat exchanger to the refrigerant vessel does not exceed a critical point.
